DEPI GRADUATION PROJECT

Automated Teacher Workflow with n8n

AI-Powered Educational Automation



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Meet Our Team

Automated Teacher Workflow with n8n

Track: AI in Education | **Team Size:** 6 Members

01 **Heba Abdelhameed Alaa Eldeen Abdelhameed Bakr**

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Company: **eyouth**

Submission Date: **1 November 2026**

1. Project background and objectives

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Project Overview

The Automated Teacher Workflow with n8n is an AI-enabled automation project that helps teachers reduce repetitive instructional content-production tasks. The teacher submits an instructional topic and related lesson information, and an n8n workflow automatically passes the data through multiple AI-powered steps to generate a lesson outline, student activities, an assessment, an answer key, and a differentiated version. The final outputs are organized for teacher review before use.

The project focuses on workflow automation rather than building a full learning platform. It demonstrates triggers, multiple workflow nodes, data passing, AI transformation, branching, conditions, error handling, a fallback path, structured output formatting, and documentation.

Problem Statement

Teachers repeatedly perform administrative and instructional content-production tasks that can be partially automated. Creating lesson outlines, activities, assessments, answer keys, and differentiated versions separately can consume time and create unnecessary repetition.

The project addresses this problem by creating one automated workflow in which the teacher provides the instructional input once, and the system processes it through a sequence of connected steps to produce multiple organized outputs for teacher review.

Project Objectives

- Automate a sequence of repetitive teacher content-production tasks using n8n.
- Accept a single instructional input and reuse it across multiple workflow steps.
- Generate a lesson outline, student activities, assessment, answer key, and differentiated version.
- Demonstrate data passing between connected workflow nodes.
- Apply AI transformation through structured prompts at relevant workflow stages.
- Use branching and conditions to control workflow behavior.
- Include input validation, error handling, and a fallback path.
- Format the final outputs in a clear structure for teacher review.
- Document the workflow, prompts, test cases, and error/fallback behavior.

Project Scope

Proposed Inputs

- Subject
- Grade level
- Topic
- Learning objective
- Lesson duration (when relevant)
- Learner level or differentiation preference (when relevant)

Core Workflow Outputs

- Lesson outline
- Student activities
- Assessment
- Answer key
- Differentiated version
- Organized final output for teacher review

Core Technical Scope

- Trigger
- Multiple workflow nodes
- Data passing between nodes
- AI transformation
- Branching
- Conditions
- Error handling
- Fallback path
- Output formatting
- Documentation

Optional Extension

- Connect the workflow to an external API or educational platform for automatic data retrieval or submission. This extension will only be considered after the core workflow is complete.

Proposed High-Level Workflow

1. Teacher submits the instructional topic and required lesson information.
2. n8n trigger starts the workflow.
3. Input validation checks that required information is available.
4. AI step generates the lesson outline.
5. AI step generates student activities.
6. AI step generates the assessment.
7. AI step generates the answer key.
8. A condition/branch applies the required differentiated version.
9. The workflow formats and combines the outputs.
10. Teacher receives the organized output for review.
11. If a node fails or required data is missing, the workflow follows an error/fallback path.

2. Task Assignment & Roles

Team Member	Primary Role	Main Responsibilities	Key Outputs
Hend Ahmed Hassan Sayed Ahmed	• Project Planning, Literature Review & Requirements	• Prepare Task 1 project planning and management; prepare Task 2 literature review; prepare Task 3 requirements gathering; coordinate documentation and final presentation.	• Tasks 1–3 documentation; planning, requirements, and presentation coordination
Eman Mahmoud Siddik Awad	• System Analysis & Design • Implementation	• Develop Task 4 system analysis and design, including architecture, data flow, diagrams, data modeling, integration design and technical design documentation.	• Task 4 system analysis and design documentation • Task 5 Implementation • Final workflow, n8n JSON, repository
Fatma Mohamed Ahmed	• System Analysis	• Support Task 4 design work and diagrams	• Task 4 system analysis
Marwa Ahmed Mohamed	• Implementation – Task 5 Parts 1 & 2	• Complete actual workflow implementation and corrections; prepare source/workflow artifacts; manage version control and collaboration evidence.	• version-control evidence
Ahmed Ismail Attia Abd Elaal	• Implementation – Task 5 Part 3: Deployment & Execution	• Prepare deployment and execution setup; verify the final workflow can run end-to-end; document execution steps and final access links.	• Deployment/execution evidence, runnable workflow, execution documentation
Heba Abdelhameed Alaa Eldeen Abdelhameed Bakr	• Testing & Quality Assurance – Task 6	• Prepare and execute the test plan and test cases; record evidence; report bugs; perform retesting after fixes; document final testing results.	• Testing report, test evidence, bug reports, and retest results

4. Risk Assessment & Mitigation Plan

Risk	Potential Impact	Mitigation Plan
Missing or unclear teacher input	The workflow may generate incomplete or unsuitable outputs.	Validate required fields before AI generation and route incomplete inputs to a clarification/fallback path.
AI output is irrelevant or inconsistent	Generated materials may not match the topic or intended lesson context.	Use structured prompts, pass the same core context between nodes, and refine prompts through repeated testing.
n8n node or integration failure	The workflow may stop before producing the final output.	Test nodes independently, use error handling, and define retry/fallback behavior.
Branching/condition logic behaves incorrectly	The wrong workflow path or output may be generated.	Define clear conditions and create dedicated test cases for every branch.
API/model limit or temporary service failure	AI steps may fail or become temporarily unavailable.	Monitor usage limits, document retry behavior, and keep a supported alternative configuration where possible.
Final outputs are poorly organized	Teacher may find the result difficult to review or use.	Use a consistent output template and review formatting in end-to-end tests.
Scope becomes larger than available time	Core workflow may remain incomplete.	Prioritize the official Project 9 core scope and postpone external API/platform integration until the core workflow works.
Team work is not integrated regularly	Late integration may create conflicts or missing documentation.	Use one GitHub repository, assign clear ownership, and review progress at short milestones.

5. Key Performance Indicators (KPIs)

KPI	How It Will Be Measured	Initial Target	Purpose
End-to-End Workflow Success Rate	Successful complete runs divided by total valid-input test runs.	≥ 90%	Measures overall workflow reliability.
Required Output Completion	Number of required outputs produced in a successful run.	100% of core outputs	Confirms that the workflow completes the intended automation sequence.
Branching & Condition Accuracy	Correct branch selected in predefined branch test cases.	100% of defined branch cases	Confirms workflow logic is functioning correctly.
Error/Fallback Handling	Defined failure and invalid-input cases handled without producing a misleading final result.	100% of defined error cases	Measures robustness of failure handling.
Content Relevance	Team/teacher-review checklist comparing generated outputs with topic, grade, objective, and input constraints.	≥ 85% acceptable results in test set	Checks usefulness and alignment of AI outputs.

KPI	How It Will Be Measured	Initial Target	Purpose
Output Formatting Completeness	Final package follows the agreed output structure with all sections clearly labeled.	100% of successful runs	Ensures teacher-facing outputs are organized and reviewable.
Processing Time Stability	Record total run time across repeated tests and compare against the established test baseline.	No unexplained major increase across repeated tests	Tracks workflow efficiency without assuming a fixed API response time.

6. Project Assumptions and Boundaries

- The project will focus on a working n8n automation and will not require building a full educational platform.
- The teacher remains responsible for reviewing AI-generated materials before classroom use.
- External API or educational-platform integration is treated as an optional extension, not a core dependency.
- The team will use one GitHub repository for project files, documentation, workflow exports, prompts, and updates.
- Any technical change that significantly expands the official Project 9 scope will be discussed with the technical instructor first.

2. Literature Review

1. Feedback & Evaluation

Project context. Task 2 continues directly from the Project Planning & Management document submitted in Task 1. The same project scope, technical direction, team structure, risks, KPIs, and project boundaries are retained unless the lecturer requests a change.

1.1 Lecturer Feedback

> **Feedback received from the lecturer (Eng. Alaa)**
>
> The lecturer stated that he has no comments on Task 1. Accordingly, no revisions to the submitted Task 1 are required at this stage, and the team will continue with the same project direction and scope.

1.2 Evaluation Summary

Evaluation Area	Current Status	Team Action
Project idea and direction	No lecturer comments on Task 1	Continue with Project #9 as submitted.
Project scope	No revision requested	Retain the core n8n workflow scope and outputs defined in Task 1.
Technical approach	No revision requested	Proceed with trigger, connected nodes, AI transformations, branching, conditions, error handling, fallback path, and formatted teacher review output.
Planning assumptions	No revision requested	Use the existing timeline, risks, KPIs, GitHub approach, and project boundaries as the implementation baseline.

2. Suggested Improvements

Because no corrective feedback was requested for Task 1, the following items are implementation-focused enhancements that strengthen reliability, consistency, and teacher usability without changing the approved project direction.

ID	Suggested Improvement	Why It Improves the Project	Priority	Implementation Stage
I1	Strengthen input validation and clarification paths	Prevents incomplete or unclear teacher inputs from reaching the AI generation steps.	**High**	Before core AI generation
I2	Use structured and reusable prompt templates	Keeps the five AI stages consistent and makes prompt refinement easier to document and test.	**High**	Prompt design and implementation
I3	Standardize data passed between workflow nodes	Reduces formatting errors and keeps subject, grade, topic, objective, and lesson context consistent across outputs.	**High**	Workflow design and build
I4	Define explicit branching rules for differentiation	Ensures the differentiated version follows clear conditions rather than an uncontrolled free-form generation step.	**High**	Branching and conditions
I5	Expand error handling, retry, and fallback behavior	Helps the workflow recover from model/API or node failures without returning a misleading final package.	**High**	Robustness and QA
I6	Create a consistent teacher-review output template	Makes the generated lesson package easier to scan, review, edit, and approve before classroom use.	**Medium**	Output formatting
I7	Use a broader test matrix before final delivery	Validates normal, edge, invalid-input, branch, and failure scenarios across more than one lesson context.	**Medium**	Testing and refinement
I8	Keep external platform/API integration optional	Protects the core deadline while leaving a clear enhancement path after the main workflow is stable.	**Optional**	After core workflow completion

3. Final Grading Criteria

> **Important note on grading weights**
>
> The DEPI documentation guidelines identify the final grading areas as Documentation, Implementation, Testing, and Presentation. The provided guideline does not state percentage weights, so this Task 2 does not invent or assign percentages.

Grading Area	What the Project Should Demonstrate	Evidence to Prepare	Current Preparation Status
Documentation	Clear and consistent project documentation that reflects the actual workflow and implementation decisions.	Task 1 planning file; Task 2; Task 3 requirements; workflow diagram; prompts; README/user guidance; GitHub documentation.	In progress
Implementation	A working n8n automation with trigger, multiple nodes, data passing, AI transformations, branching/conditions, error handling, fallback path, and organized outputs.	Working workflow; exported n8n JSON; prompt versions; sample generated lesson package.	Planned for implementation phase
Testing	Evidence that the workflow was tested under normal, edge, invalid-input, branch, and failure scenarios, with issues corrected.	Test cases and results; error/fallback demonstration; issue and fix log; final regression run.	Planned for testing phase
Presentation	A clear explanation and demonstration of the problem, solution, workflow logic, implementation choices, challenges, and project outcomes.	Final PPT/PDF; live or recorded demo; concise explanation of team roles and project results.	Planned for final delivery

3.1 Task 2 Outcome

- Lecturer feedback has been documented: no comments were provided on Task 1.
- No changes are required to the Task 1 project direction at this stage.
- The team will use the suggested improvements above as implementation-quality enhancements, not as changes to the approved scope.
- The project will continue to Requirements Gathering and System Analysis & Design using the Task 1 baseline.

3. Requirements Gathering

1. Requirements Gathering Overview

This document defines who the project serves, how users will interact with the workflow, and what the system must do. It is based on the approved project scope from Task 1: a teacher submits instructional information and an n8n workflow generates a lesson outline, student activities, an assessment, an answer key, and a differentiated version, then organizes the outputs for teacher review.

Core Requirement Boundary
The core project is an n8n automation, not a full LMS or student-facing platform. External API or educational-platform integration remains optional until the core workflow is stable.

1.1 Requirements Context View

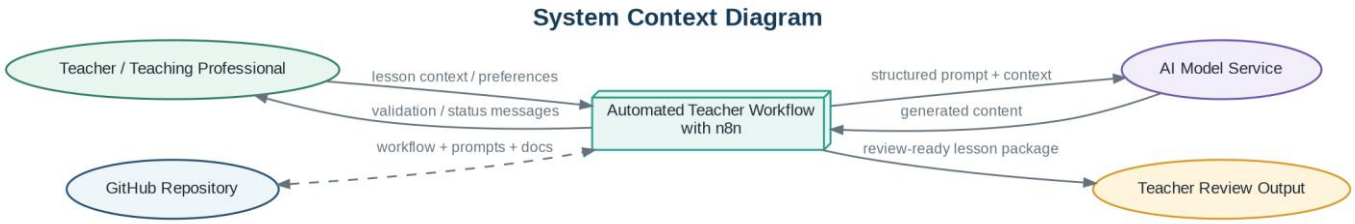


Figure 1. Requirements Context Diagram.

2. Stakeholder Analysis

Stakeholder	Role	Interaction	Key Needs	Influence	Interest
Teacher / Teaching Professional	Primary user	Enter lesson information; receive and review generated materials.	Fast preparation, clear outputs, relevant content, control over inputs, understandable errors.	High	High
Students	Indirect beneficiary	Benefit from activities, assessments, and differentiated materials after teacher review.	Age/level-appropriate materials, clarity, appropriate differentiation.	Medium	Medium
Technical Instructor / Project Reviewer	Evaluator / advisor	Review scope, requirements, implementation evidence, and project quality.	Clear documentation, traceable requirements, functioning workflow, justified decisions.	High	High
Project Team	Builder / maintainer	Design, build, test, document, and maintain the workflow.	Clear scope, modular logic, version control, testable requirements, manageable workload.	High	High

2.1 Stakeholder Priorities

- Teacher needs are the primary driver of the functional requirements because teachers are the direct users of the workflow.
- Student needs influence the quality and differentiation of generated materials, but students do not directly operate the core system.
- The Technical Instructor and reviewers need evidence that the workflow satisfies the documented scope and can be tested and explained.
- The project team needs requirements that are clear enough to divide work, build the workflow, and verify completion.

3. User Stories

ID	Actor	User Story	Acceptance Criteria
US-01	Teacher	As a teacher, I want to submit the lesson topic and required instructional information so that the workflow can start from one structured input.	Required fields are accepted and the workflow starts only when mandatory inputs are present.
US-02	Teacher	As a teacher, I want the system to generate a lesson outline so that I can prepare the lesson structure faster.	A clearly labeled lesson outline is produced using the submitted context.
US-03	Teacher	As a teacher, I want student activities generated from the same lesson context so that the activities remain relevant to the lesson.	Activities reflect the topic, grade/level, and learning objective supplied.
US-04	Teacher	As a teacher, I want an assessment and answer key generated together so that I can review questions and correct answers efficiently.	Assessment items and a corresponding answer key are both produced and clearly separated.
US-05	Teacher	As a teacher, I want a differentiated version so that the material can better match the selected learner level or differentiation preference.	The workflow follows the correct branch and produces the requested differentiated output.
US-06	Teacher	As a teacher, I want all generated outputs organized in one clear package so that I can review them before classroom use.	All core outputs are labeled, ordered, and presented for teacher review.
US-07	Teacher	As a teacher, I want a clear message when my input is incomplete or the workflow fails so that I know what to correct or retry.	Invalid input or execution failure is routed to an understandable error/fallback path.
US-08	Technical Instructor / Reviewer	As a reviewer, I want the workflow requirements and behavior documented so that I can verify the project against its intended scope.	Requirements, use cases, workflow behavior, and testable criteria are documented and traceable.
US-09	Project Team	As a project team, we want prompts and workflow logic to be maintainable and versioned so that changes can be tracked during implementation and testing.	Prompt versions and workflow changes are stored and documented in the project repository.

4. Use Cases

4.1 Use Case Diagram

Visual summary of the documented input, validation, generation, differentiation, review, and fallback use cases.

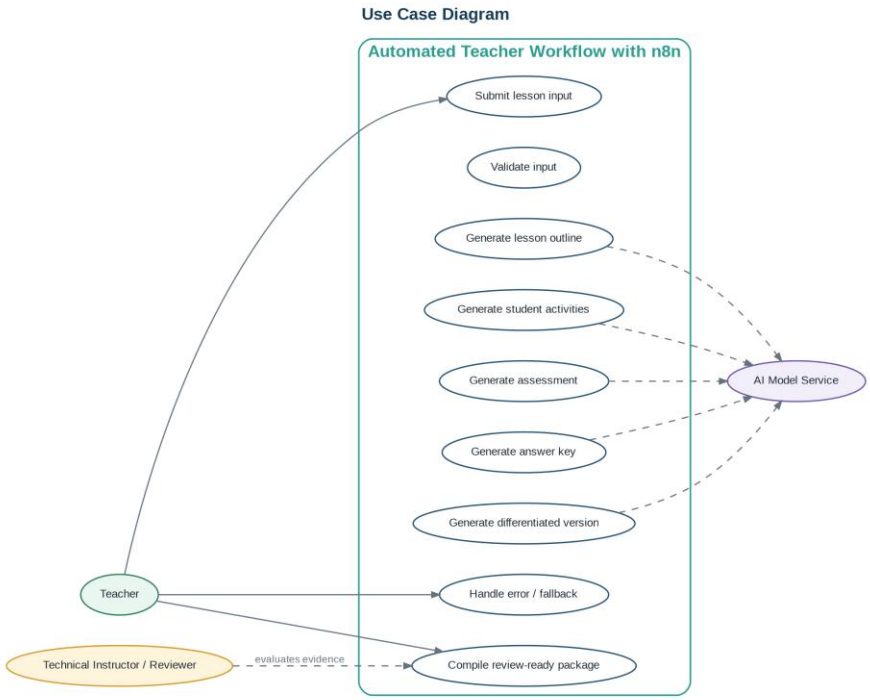


Figure 2. Requirements Use Case Diagram - primary actor interactions and core workflow use cases.

ID	Use Case	Actor	Precondition	Main Flow	Outcome
UC-01	Submit instructional input	Teacher	Teacher has access to the input form/trigger.	Teacher submits subject, grade level, topic, learning objective, and relevant options.	Input is received by n8n.
UC-02	Validate input	System / Teacher	UC-01 completed.	Workflow checks required fields and basic validity.	Valid input continues; invalid input goes to clarification/fallback.
UC-03	Generate lesson outline	Teacher / AI workflow	Validated input is available.	AI receives structured context and creates the lesson outline.	Lesson outline output is produced.
UC-04	Generate student activities	Teacher / AI workflow	Shared lesson context is available.	AI generates student activities using the same context.	Activity output is produced.
UC-05	Generate assessment	Teacher / AI workflow	Shared lesson context is available.	AI generates assessment items according to the lesson context.	Assessment output is produced.
UC-06	Generate answer key	Teacher / AI workflow	Assessment output is available.	AI produces a corresponding answer key.	Answer key is produced and associated with the assessment.
UC-07	Generate differentiated version	Teacher / Workflow branch	Differentiation input/condition is available.	Condition routes the workflow to the appropriate differentiation step.	Differentiated version is produced.
UC-08	Compile output for review	Teacher	Core generation steps complete.	Workflow combines and labels all outputs in a consistent structure.	Teacher receives the review-ready package.
UC-09	Handle failure or invalid input	Teacher / System	A validation or execution problem occurs.	Workflow records/identifies the issue and routes to error handling or fallback behavior.	Teacher receives a clear corrective or retry message instead of a misleading final result.

4.2 Main Success Scenario

- Teacher submits valid instructional information.
- n8n trigger receives the data and validates the required fields.
- The workflow passes the same core lesson context through the AI generation stages.
- Lesson outline, activities, assessment, answer key, and differentiated version are generated.
- The outputs are formatted, combined, and presented for teacher review.
- The teacher reviews the material before any classroom use.

4.3 Alternative / Failure Scenario

- If required input is missing, the workflow does not continue to generation and asks for correction.
- If an AI/API step fails or returns unusable output, the workflow uses the defined retry or fallback behavior.
- The workflow should not present an incomplete or misleading package as a successful final result.

4.4 Requirements Activity Flow

Visual flow of validation, generation, differentiation, fallback, and teacher review across the documented scenarios.

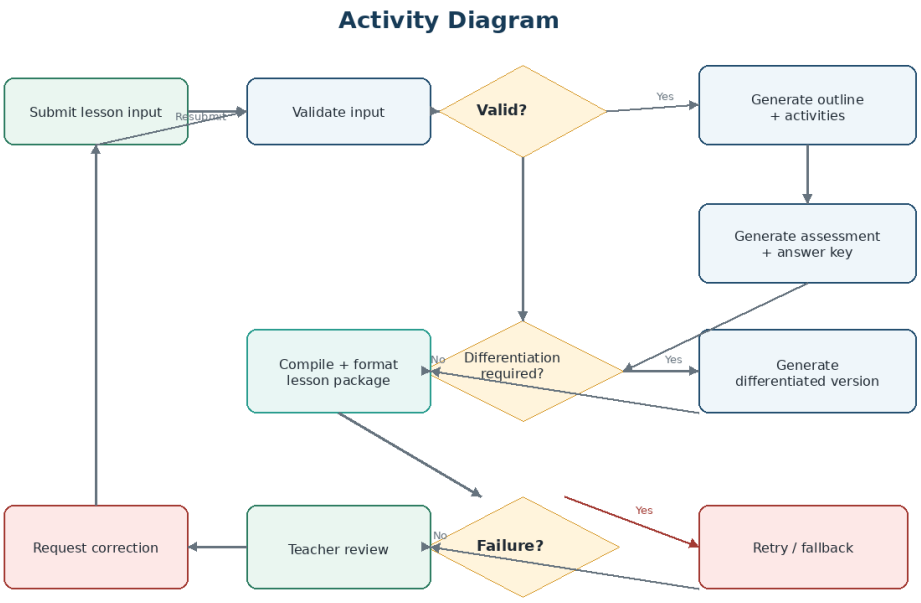


Figure 3. Requirements Activity Flow - validation, generation, differentiation, fallback, and review.

4.5 Requirements Data Flow Models

These models visualize how the documented teacher input and generated outputs move through the system. They support the functional requirements without replacing the detailed design documentation in Task 4.

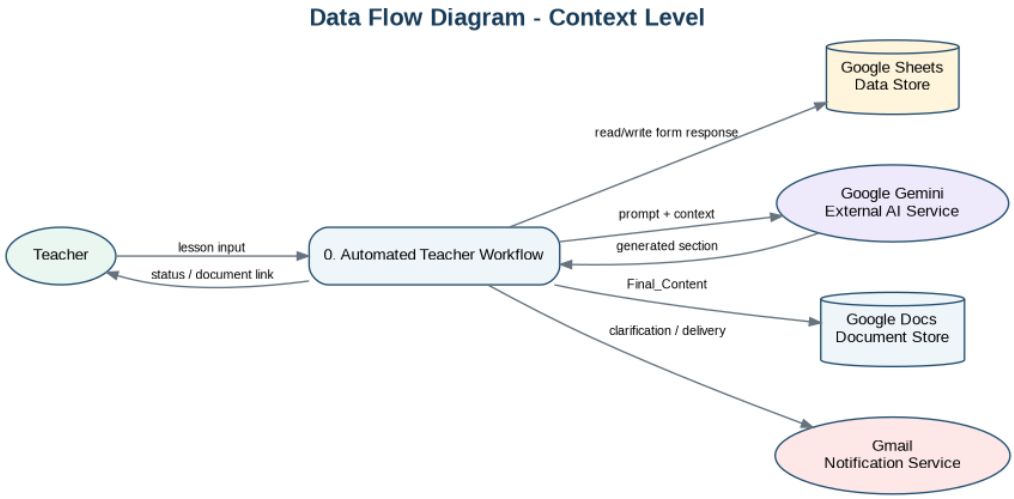


Figure 4. Requirements Data Flow - Context Level.

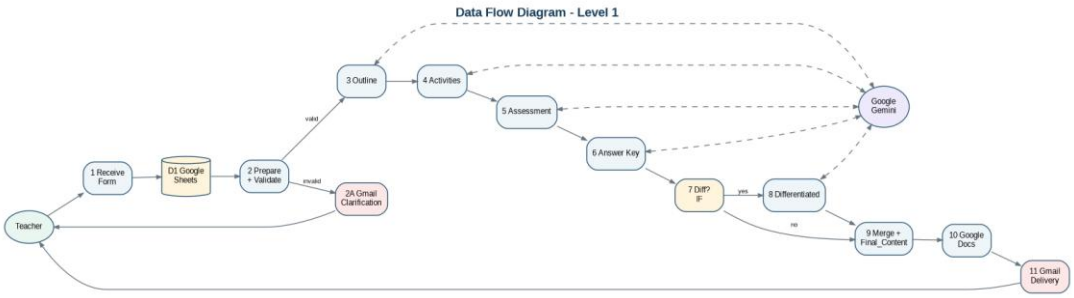


Figure 5. Requirements Data Flow - Level 1.

4.6 Requirements Interaction Sequence

The sequence view shows the expected order of interaction between the teacher, input services, the n8n workflow, AI generation, document creation, and communication services.

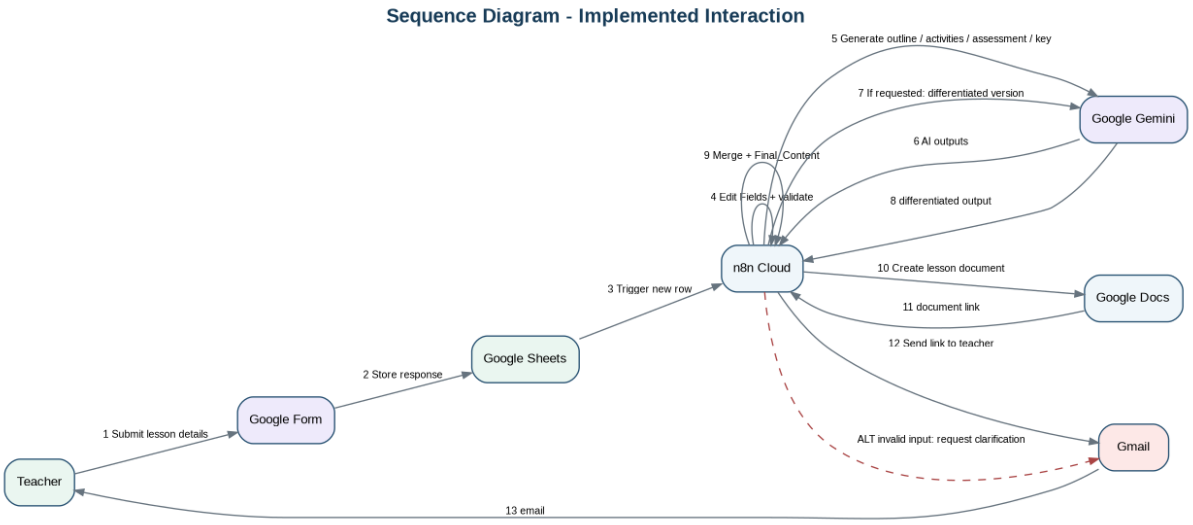


Figure 6. Requirements Interaction Sequence.

4.7 Requirements State Model

The state model summarizes the expected workflow states from input receipt through validation, generation, differentiation, output creation, error handling, and completion.

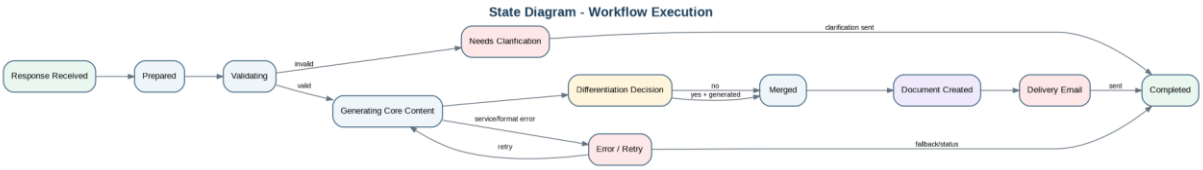


Figure 7. Requirements State Model.

5. Functional Requirements

ID	Requirement	Priority
FR-01	The system shall provide a trigger/form to receive teacher instructional input.	Must
FR-02	The system shall accept subject, grade level, topic, and learning objective as core lesson context.	Must
FR-03	The system shall accept lesson duration and differentiation/learner-level preference when relevant.	Should
FR-04	The workflow shall validate mandatory fields before starting AI generation.	Must
FR-05	The workflow shall pass the required shared context between relevant nodes.	Must
FR-06	The workflow shall generate a lesson outline.	Must
FR-07	The workflow shall generate student activities.	Must
FR-08	The workflow shall generate an assessment.	Must
FR-09	The workflow shall generate an answer key corresponding to the assessment.	Must
FR-10	The workflow shall generate a differentiated version using defined conditions/branching.	Must
FR-11	The workflow shall use clear prompt instructions and structured output formats for AI stages.	Must
FR-12	The workflow shall combine and label all core outputs for teacher review.	Must
FR-13	The workflow shall include conditions and branching where the selected input changes workflow behavior.	Must
FR-14	The workflow shall include error handling for failed nodes or AI/API execution problems.	Must
FR-15	The workflow shall provide a fallback/corrective path for invalid input or failed generation.	Must
FR-16	The project shall retain prompts, workflow files/exports, and documentation in the project repository.	Must
FR-17	The teacher shall review generated outputs before classroom use.	Must
FR-18	The workflow may connect to an external API or educational platform after core implementation is stable.	Could

6. Non-Functional Requirements

ID	Category	Requirement	Verification
NFR-01	Usability	The teacher-facing input and output structure should be clear, labeled, and understandable without requiring programming knowledge.	Teacher can identify required inputs and review each generated output section.
NFR-02	Reliability	The workflow should complete successfully for at least 90% of valid-input test runs, aligned with the Task 1 KPI.	Measured from test execution history.
NFR-03	Output Completeness	Every successful core run should produce all required core outputs.	100% of core outputs present in successful runs.
NFR-04	Branch Accuracy	Defined branch/condition test cases should route to the correct workflow path.	100% of defined branch cases during testing.
NFR-05	Error Handling	Defined invalid-input and simulated failure cases should be handled without presenting a misleading final result.	100% of defined error cases during testing.
NFR-06	Content Relevance	Generated outputs should remain relevant to the submitted topic, grade/level, learning objective, and constraints.	At least 85% acceptable results in the agreed review test set.
NFR-07	Performance	Processing time should remain reasonably stable across repeated tests with comparable inputs.	Execution time recorded and compared with the established test baseline.
NFR-08	Security	API keys and credentials must not be hard-coded into shared prompts, documents, or exported public files.	Credentials stored using n8n credential/configuration mechanisms.
NFR-09	Privacy	The core workflow should avoid unnecessary personal student data because Project #9 can operate on lesson-level inputs.	No student personal data required for normal core use.
NFR-10	Maintainability	Prompt templates and workflow logic should be organized, documented, and versioned for team maintenance.	Meaningful filenames/versions and repository updates.
NFR-11	Compatibility	The workflow should run in the team's selected n8n environment and use supported AI/API integrations available to the team.	Verified during environment setup and implementation.

7. Requirements Priority Summary

Priority	Scope
Must Have	Input trigger and validation; five core outputs; data passing; AI transformations; branching/conditions; error handling; fallback path; output formatting; teacher review; documentation.
Should Have	Lesson duration and differentiation controls where relevant; refined prompt structures; clearer user messages and execution logging.
Could Have	External API or educational-platform integration after the core workflow is stable.
Not in Core Scope	Full LMS/student-facing platform, automatic grading of student submissions, or custom AI model training/fine-tuning.

8. Requirements Traceability

User Story	Mapped Functional Requirements	Verification Focus
US-01	FR-01, FR-02, FR-03, FR-04	Input received and validated.
US-02	FR-05, FR-06, FR-11	Lesson outline generated from shared context.
US-03	FR-05, FR-07, FR-11	Student activities generated from shared context.
US-04	FR-08, FR-09, FR-11	Assessment and answer key generated and linked.
US-05	FR-10, FR-13	Correct differentiated branch and output.
US-06	FR-12, FR-17	Complete organized package available for teacher review.
US-07	FR-04, FR-14, FR-15	Invalid/failure case follows clear error or fallback path.
US-08	FR-16	Requirements and workflow evidence documented for review.
US-09	FR-16	Prompts/workflow changes stored and maintainable in repository.

8.1 Requirement Verification Coverage

This coverage view links key requirement areas to the evidence expected during implementation and testing, supporting traceability from documented requirements to verification.

Testing & Validation Coverage				
Area	**Input/Trigger**	**AI Stages**	**Branching**	**Output/Delivery**
Node / Unit	Validate	Prompt + response	IF nodes	Merge / Docs / Gmail
Integration	Forms→Sheets→n8n	n8n↔Gemini	Diff yes/no	Docs + email link
Failure	Missing fields	Model/API error	Invalid preference	Doc/email failure
User Review	Teacher can open, review and edit the generated Google Docs lesson plan			

Figure 8. Requirement Verification Coverage.

9. Assumptions & Constraints

- The core user is a teacher or teaching professional; students benefit indirectly and do not need direct access in the core project.
- Teacher review is required before AI-generated materials are used.
- The team will implement the core n8n workflow before attempting optional external integration.
- The workflow depends on an available n8n environment and a supported AI model/API configuration.
- No official requirement in the current project scope requires a database, full front-end application, or LMS.
- Requirements may be refined after Technical Instructor feedback, but scope changes should be documented and agreed by the team.

4.System Analysis and Design

This document reflects the implemented solution: Google Forms for teacher input, Google Sheets for response capture/logging, n8n Cloud for orchestration, Google Gemini for AI generation, Google Docs for the editable lesson plan, and Gmail for clarification/delivery.

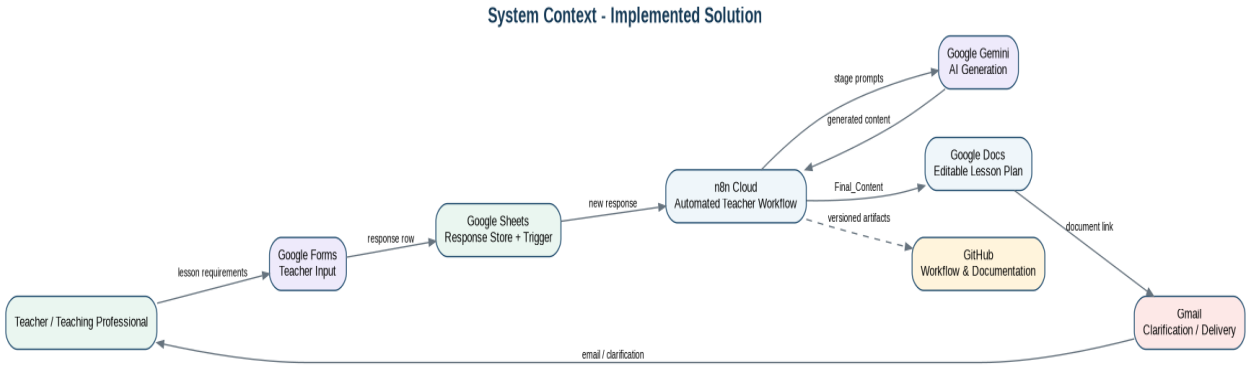


Figure 1. System Context Diagram - implemented Google Workspace, n8n Cloud, Gemini, Docs, Gmail, and repository context.

1. Problem Statement & Objectives

1.1 Problem Statement

The Automated Teacher Workflow streamlines lesson preparation by collecting teacher requirements through Google Forms, storing the response in Google Sheets, and using an n8n Cloud workflow to prepare and validate the input. Google Gemini generates the lesson outline, student activities, assessment, answer key, and differentiated material when requested. The workflow then merges the outputs into Final_Content, creates an editable Google Docs lesson plan, and supports teacher communication through Gmail. This reduces repetitive planning work while keeping the teacher responsible for reviewing the final material.

1.2 Project Objectives

- Collect structured lesson requirements through Google Forms and capture them in Google Sheets.
- Automatically trigger the n8n Cloud workflow when a new form response is available.
- Validate and prepare the submitted lesson data before AI generation begins.
- Use Google Gemini to generate the lesson outline, student activities, assessment, answer key, and differentiated material when requested.
- Merge the generated sections into one Final_Content field and create an editable Google Docs lesson plan.
- Use Gmail to request clarification for missing required inputs and to support delivery of the generated document link.

1.3 System Boundary

In Scope	Out of Scope
Google Forms teacher input; Google Sheets response storage/trigger; n8n Cloud orchestration; Google Gemini content generation; IF validation and differentiation branching; Merge/Edit Fields; Google Docs document creation; Gmail clarification/delivery; Google Sheets logging.	Full LMS; direct student-facing application; automatic grading of student submissions; training or fine-tuning a custom AI model.
Editable lesson-plan output; teacher review before use; workflow documentation, diagrams, prompts, and testing evidence.	Automatic classroom publishing or use of generated content without teacher review.

2. Use Case Diagram & Descriptions

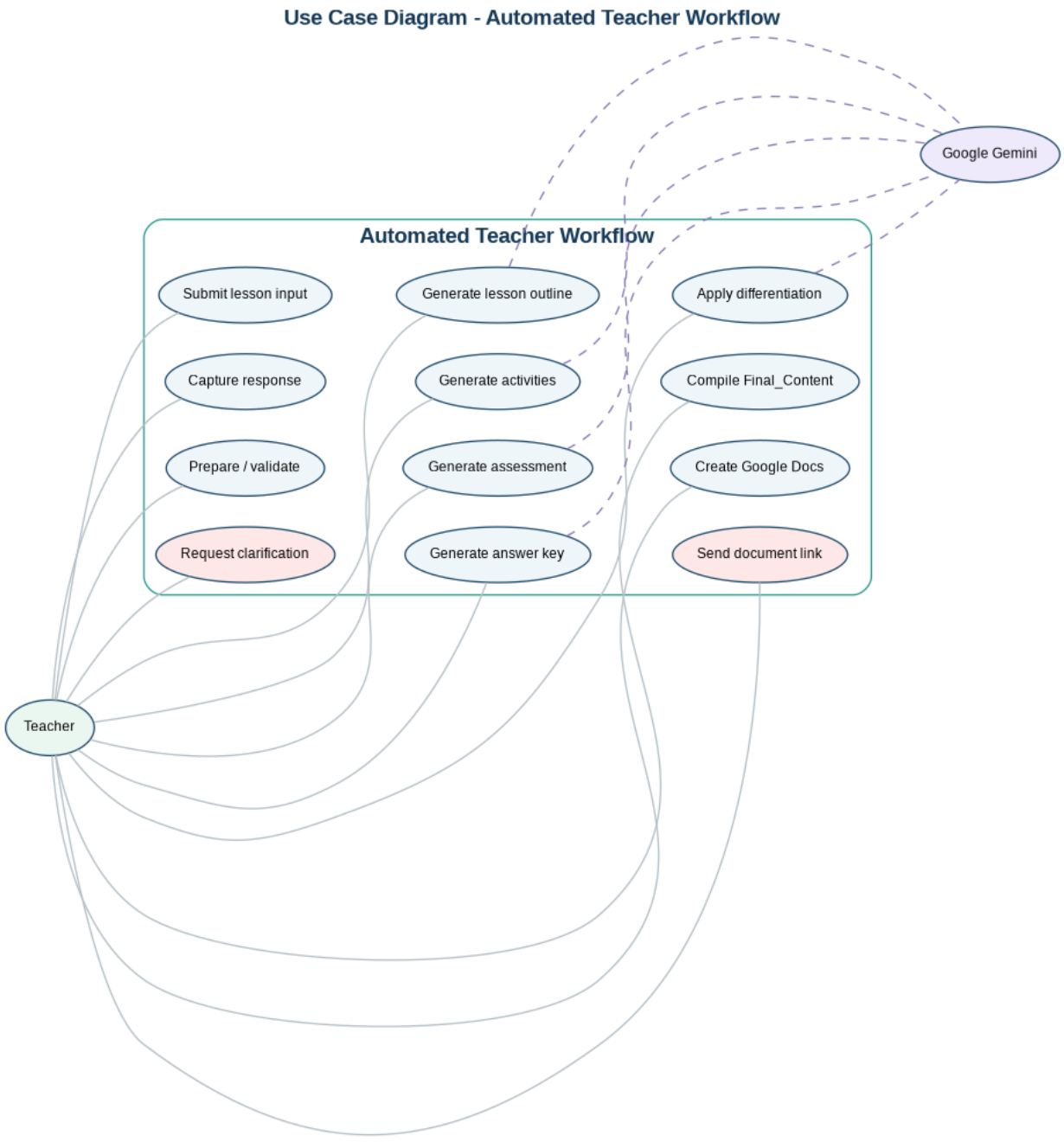


Figure 2. Use Case Diagram - teacher interactions and implemented workflow services.

ID	Use Case	Actor	Description	Result
UC-01	Submit Instructional Input	Teacher	Submit subject, grade level, topic, learning objective, lesson duration, and differentiation preference through Google Forms.	Form response is stored in Google Sheets.
UC-02	Capture Response & Trigger	System	Google Sheets receives the form response and the Google Sheets Trigger starts the n8n workflow.	New response enters n8n.
UC-03	Prepare & Validate Input	System	Edit Fields standardizes the values and an IF node checks required fields.	Valid input continues; invalid input is routed for clarification.
UC-04	Request Clarification	System / Teacher	Gmail sends a clarification request when required fields are missing.	Teacher receives a correction request.
UC-05	Generate Lesson Outline	AI Workflow	Google Gemini generates a structured lesson outline from the prepared context.	Lesson outline produced.
UC-06	Generate Student Activities	AI Workflow	Google Gemini generates activities aligned with the lesson requirements.	Student activities produced.
UC-07	Generate Assessment	AI Workflow	Google Gemini generates assessment questions aligned with the lesson context.	Assessment produced.
UC-08	Generate Answer Key	AI Workflow	Google Gemini generates answers corresponding to the assessment.	Answer key produced.
UC-09	Apply Differentiation	Workflow Branch / AI	An IF node checks the teacher preference; Google Gemini generates differentiated material when requested.	Differentiated material included when applicable.
UC-10	Compile Final Content	System	Merge combines the relevant outputs and Edit Fields builds the Final_Content field.	One combined lesson package is prepared.
UC-11	Create Google Docs Lesson Plan	System	Google Docs creates an editable lesson-planning document containing Final_Content.	Editable lesson document created.
UC-12	Deliver Document Link	System / Teacher	Gmail sends or is configured to send the generated document link and summary to the teacher.	Teacher can access the generated document.

3. Functional & Non-Functional Requirements

3.1 Functional Requirements

ID	Functional Requirement
FR-01	Collect teacher lesson information through Google Forms.
FR-02	Store the submitted form response in Google Sheets and use the response to trigger the n8n workflow.
FR-03	Use Edit Fields to standardize and prepare the submitted values for downstream nodes.
FR-04	Validate required input fields with an IF node before AI generation begins.
FR-05	Send a Gmail clarification request when mandatory input is missing.
FR-06	Generate a lesson outline using Google Gemini.
FR-07	Generate student activities using Google Gemini.
FR-08	Generate assessment questions using Google Gemini.
FR-09	Generate a corresponding answer key using Google Gemini.
FR-10	Use an IF branch to determine whether differentiated material is required.
FR-11	Generate a differentiated version with Google Gemini when the teacher requests it.
FR-12	Merge the relevant generated outputs into a single workflow path.

ID	Functional Requirement
FR-13	Assemble the combined sections into one Final_Content field using Edit Fields.
FR-14	Create an editable Google Docs lesson-plan document from Final_Content.
FR-15	Append/store teacher input data in Google Sheets for record/logging purposes.
FR-16	Support Gmail delivery of the generated Google Docs link to the teacher.
FR-17	Maintain the workflow and supporting project documentation in the project repository.

3.2 Non-Functional Requirements

ID	Category	Requirement
NFR-01	Usability	The teacher input and final lesson plan must be clearly structured and understandable without programming knowledge.
NFR-02	Reliability	The workflow should complete successfully for valid form submissions and handle invalid input through the clarification path.
NFR-03	Completeness	A successful run should generate the required lesson sections and combine them into Final_Content.
NFR-04	Branch Accuracy	The differentiation IF node should select the correct path based on the teacher preference.
NFR-05	Error Handling	Missing required input should not continue to AI generation; it should produce a clear clarification message.
NFR-06	Content Relevance	Generated content should remain aligned with subject, grade level, topic, learning objective, duration, and differentiation preference.
NFR-07	Editability	The generated lesson plan should remain editable in Google Docs so the teacher can review and adapt it.
NFR-08	Security	Google and AI credentials must be stored in n8n credentials/configuration and not embedded in shared workflow files or documentation.
NFR-09	Privacy	The workflow should process only the lesson information required for generation and avoid unnecessary personal/student data.
NFR-10	Maintainability	Workflow nodes, prompts, field names, and documentation should be organized and consistently named for future updates.
NFR-11	Integration	The n8n workflow should use supported Google Workspace and Google Gemini connectors available in the project environment.

4. Software Architecture

The implemented solution uses a cloud-based workflow architecture. Google Forms collects teacher lesson requirements and Google Sheets stores the response. A Google Sheets Trigger starts the n8n Cloud workflow, where Edit Fields prepares the values and IF nodes control validation and differentiation. Google Gemini Chat Model nodes generate the educational content. Merge and Edit Fields assemble Final_Content, Google Docs creates the editable lesson plan, and Gmail supports clarification and document-link delivery.

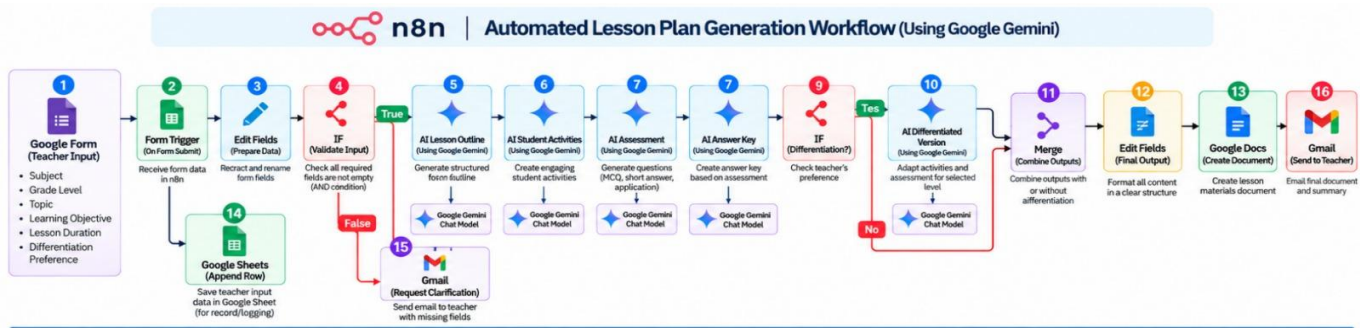


Figure 3. Implemented n8n workflow overview supplied by the project team.

Layer	Component	Responsibility
Presentation / Input	Google Forms	Collect teacher lesson requirements.
Data Capture / Trigger	Google Sheets + Google Sheets Trigger	Store form responses and start the n8n workflow when a new response is available.
Workflow Orchestration	n8n Cloud	Prepare data, validate inputs, route conditions, call AI nodes, merge outputs, and coordinate integrations.
AI Generation	Google Gemini Chat Model	Generate lesson outline, student activities, assessment, answer key, and differentiated material.
Output / Document	Google Docs	Create and store the combined editable lesson-plan document.
Notification	Gmail	Request clarification for missing fields and support delivery of the generated document link.
Version Control	GitHub / project repository	Store workflow exports, prompts, diagrams, documentation, and revisions.

5. Database Design & Data Modeling

The implemented workflow does not use a separate relational database. Google Sheets acts as the lightweight response and logging store, while n8n carries prepared fields and AI outputs as runtime JSON/item data. The final combined content is stored as an editable Google Docs document. The conceptual data model below documents the relationships among these implemented data objects without claiming a relational database that is not part of the current solution.

5.1 ER Diagram

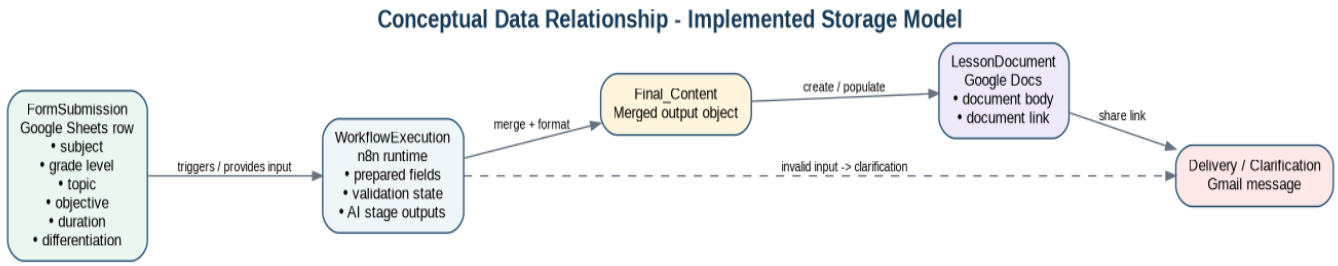


Figure 4. Conceptual Data Relationship Diagram - Google Sheets response, n8n runtime, Final_Content, Google Docs, and Gmail.

5.2 Logical Schema

Data Object	Key Fields / Content	Storage / Purpose
FormSubmission	subject; grade_level; topic; learning_objective; lesson_duration; differentiation_preference; response timestamp	Google Sheets response row; source data for the workflow.
PreparedInput	standardized field names and cleaned values	n8n runtime item produced by Edit Fields.
AIStageOutput	lesson_outline; student_activities; assessment; answer_key; differentiated_version (when requested)	n8n runtime outputs returned by Google Gemini nodes.
FinalContent	Final_Content combined text	n8n merged/formatted payload used to populate Google Docs.
LessonDocument	document body; Google Docs document reference/link	Google Docs; editable teacher-facing lesson plan.
EmailMessage	clarification message or document-link delivery payload	Gmail; teacher communication.

5.3 Physical Schema

Physical Store / Structure	Implemented Fields / Representation	Keys & Constraints
Google Sheets response row	Subject; Grade Level; Topic; Learning Objective; Lesson Duration; Differentiation Preference; response timestamp	No relational PK/FK; one Google Forms response is stored as one sheet row.
n8n prepared item	Normalized lesson fields after Edit Fields	Field names are kept consistent across downstream nodes.
n8n AI outputs	Lesson outline; Student activities; Assessment; Answer key; Differentiated version when applicable	Outputs are generated as node data and passed to Merge.
Final_Content	Combined and formatted lesson sections created by Edit Fields1	Single final field used as the Google Docs document body.
Google Docs document	Editable lesson-plan content + generated document link/reference	Created from Final_Content for teacher review.
Gmail payload	Clarification or delivery subject/body + teacher recipient + document link where applicable	Uses Gmail credential in n8n; exact recipient source depends on the form/workflow configuration.

5.4 Normalization Considerations

- One Google Forms response is represented by one Google Sheets row to keep the submitted lesson context together.
- Edit Fields standardizes names before AI processing so the same context can be reused across all Google Gemini stages.
- AI outputs remain separate until the Merge node, preventing one stage from overwriting another stage output.
- Final_Content is created only after the relevant branches are complete, then stored in Google Docs as the editable final artifact.

6. Data Flow & System Behavior

6.1 DFD - Context Level

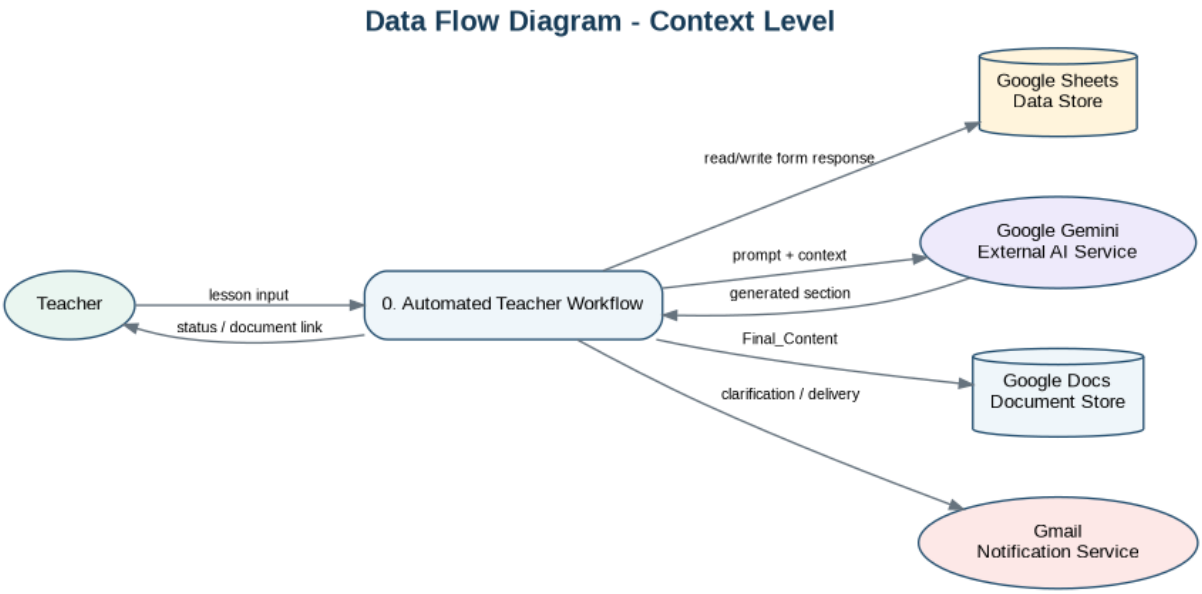


Figure 5. DFD Context Level - external services and teacher data flows.

6.2 DFD - Level 1

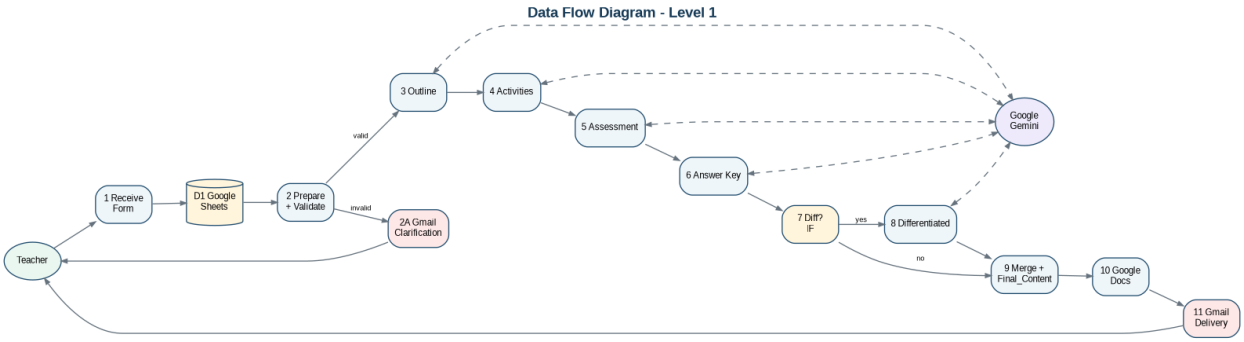


Figure 6. DFD Level 1 - implemented generation, validation, differentiation, merge, document, and delivery flow.

6.3 Sequence Diagram

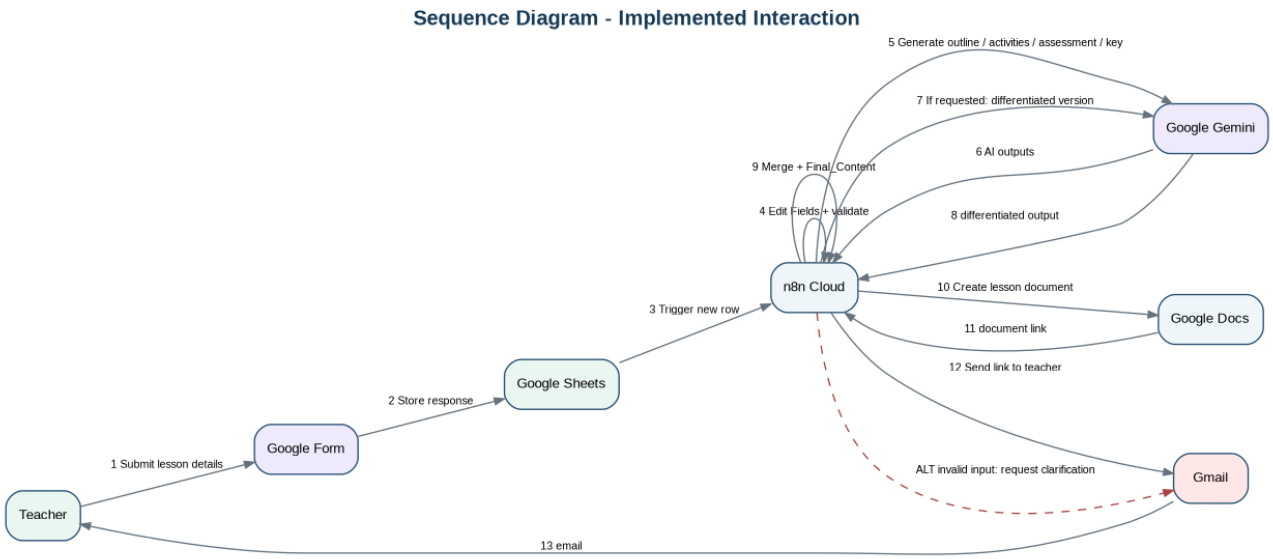


Figure 7. Sequence Diagram - Google Form submission through Google Docs and Gmail delivery.

6.4 Activity Diagram

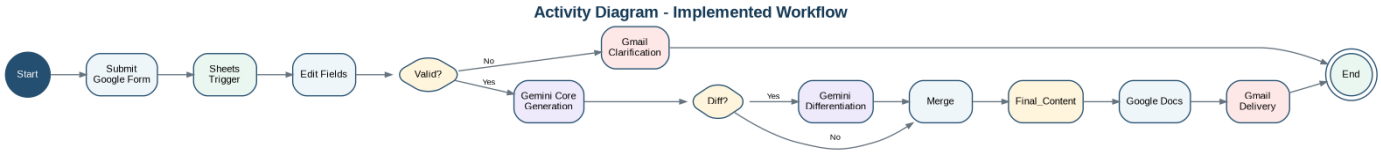


Figure 8. Activity Diagram - implemented validation, generation, branching, merge, and output flow.

6.5 State Diagram

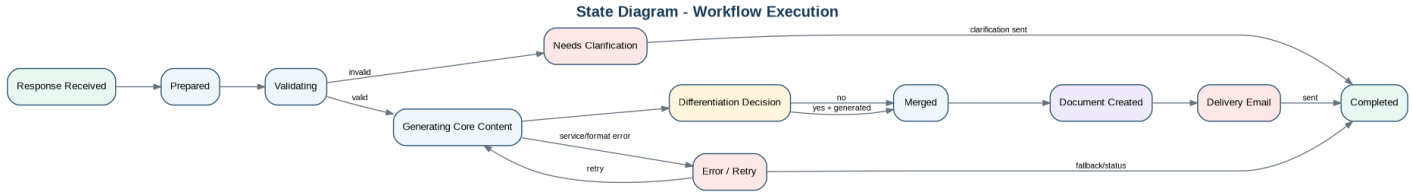


Figure 9. State Diagram - workflow execution lifecycle.

6.6 Class Diagram

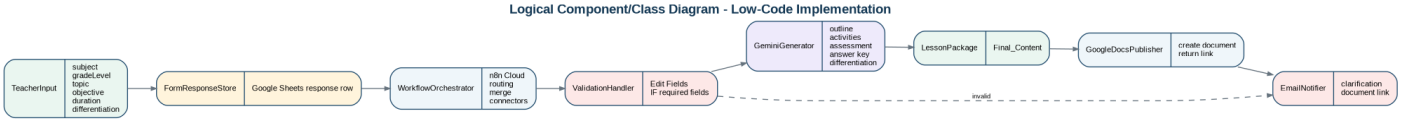


Figure 10. Logical Component/Class Diagram - low-code implementation responsibilities.

Logical Component	Responsibility
TeacherInput	Represents the lesson requirements submitted through Google Forms.
FormResponseStore	Stores the Google Forms response in Google Sheets and exposes it to the trigger.
WorkflowOrchestrator	Coordinates n8n nodes, routing, validation, AI calls, merge, and external services.
ValidationHandler	Uses Edit Fields and IF logic to prepare data and stop incomplete requests.
GeminiGenerator	Produces stage-specific lesson content using Google Gemini.
LessonPackage	Represents the merged Final_Content output.
GoogleDocsPublisher	Creates the editable lesson-plan document from Final_Content.
EmailNotifier	Sends clarification or document-link messages through Gmail.

7. UI/UX Design & Prototyping

7.1 Wireframes & Mockups

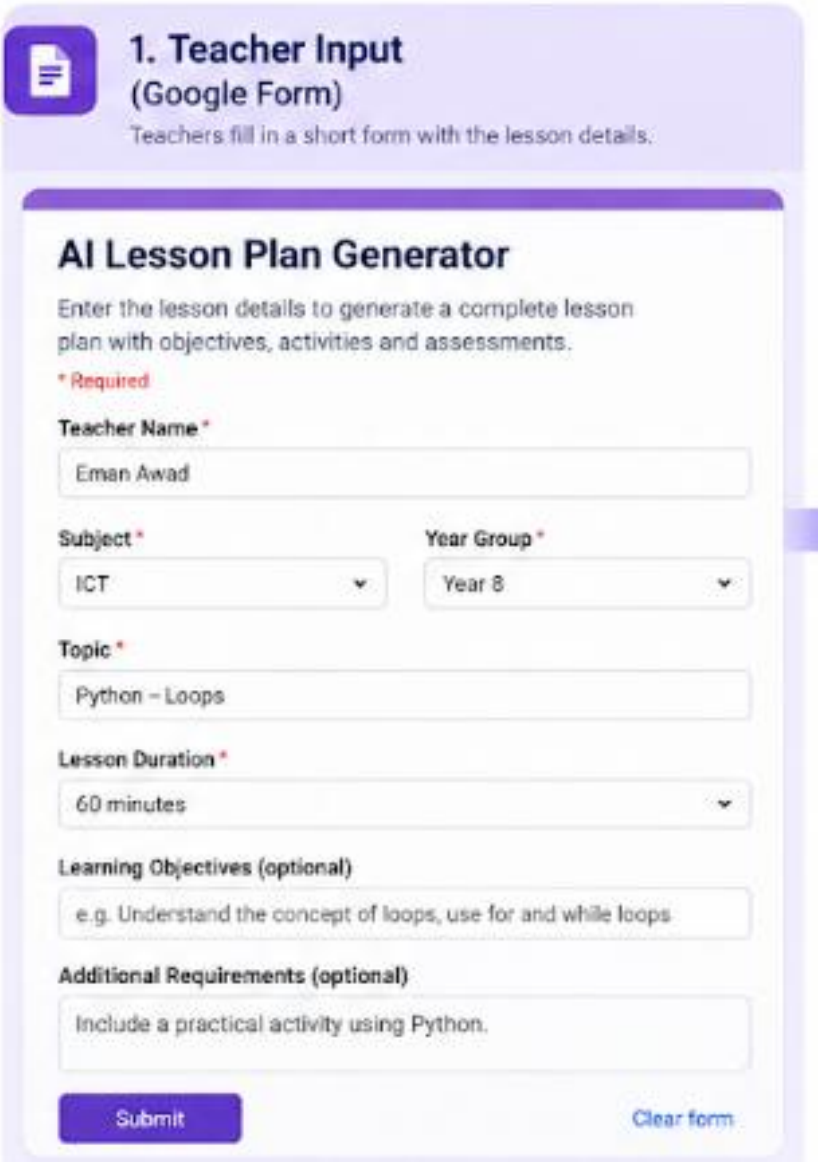


Figure 11. Teacher Input Form - Google Forms implementation visual supplied by the project team.



Figure 12. Teacher Review Output - generated Google Docs lesson-plan visual supplied by the project team.

7.2 UI/UX Guidelines

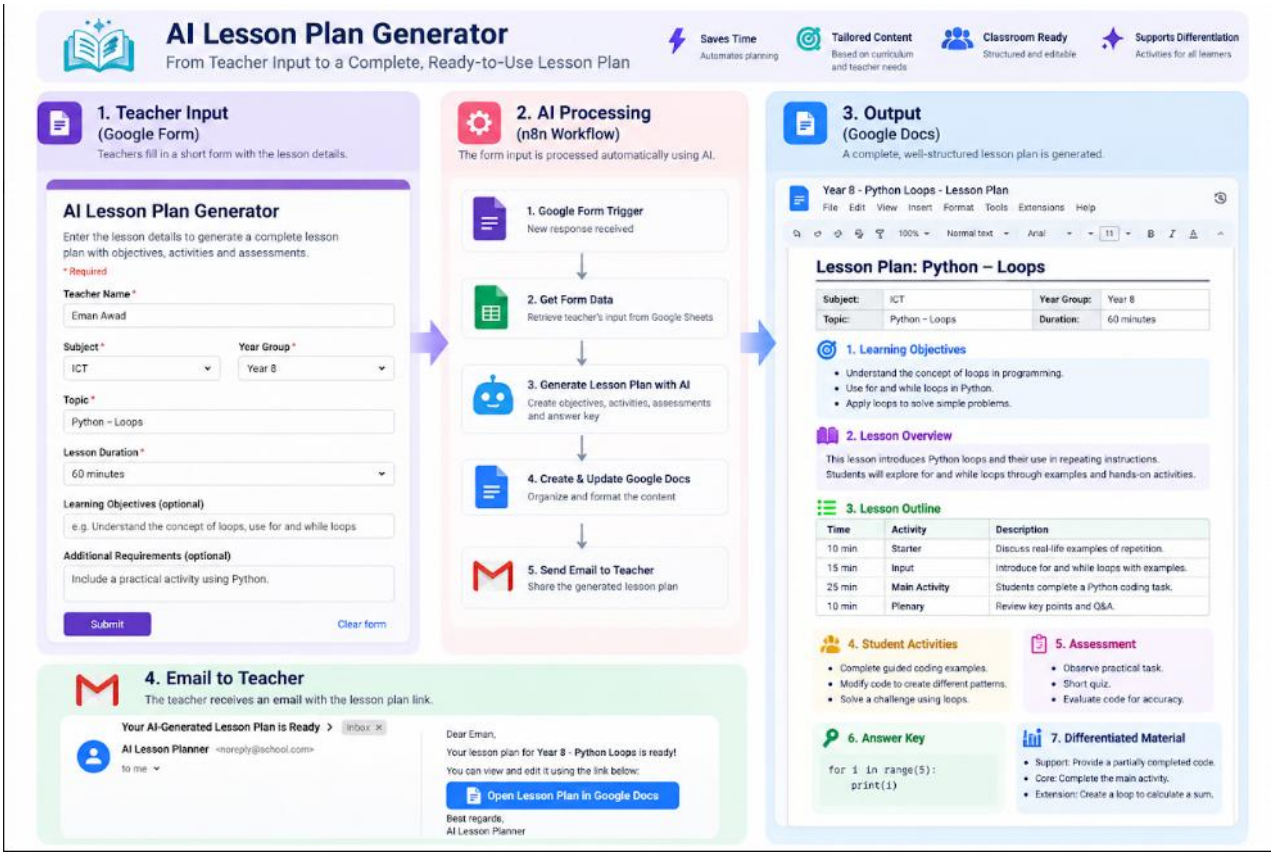


Figure 13. End-to-end teacher experience - Google Form, n8n/AI processing, Google Docs output, and email delivery.

Principle	Guideline
Clarity	Keep Google Form labels short and explicit for subject, grade level, topic, objective, duration, and differentiation preference.
Consistency	Use the same field names from Google Forms/Sheets through Edit Fields and AI prompts.
Validation	Route incomplete submissions to a specific clarification email instead of continuing generation.
Review-first	Present the output as an editable Google Docs draft that the teacher reviews before use.
Structure	Use a consistent lesson-plan hierarchy for outline, activities, assessment, answer key, and differentiated material.
Accessibility	Use readable text, clear headings, sufficient contrast, and avoid status communication through color alone.
Editability	Keep the final output in Google Docs so teachers can revise and adapt the generated content.
Feedback	Communicate missing-input or delivery status through clear Gmail messages where applicable.

8. System Deployment & Integration

8.1 Technology Stack

Layer	Technology	Purpose
Teacher Input	Google Forms	Collect subject, grade level, topic, learning objective, lesson duration, and differentiation preference.
Response Store / Trigger	Google Sheets + Google Sheets Trigger	Store form responses and trigger n8n on a new submission.
Backend / Orchestration	n8n Cloud	Run validation, AI stages, branching, merging, formatting, and service integrations.
AI Layer	Google Gemini Chat Model	Generate the lesson outline, activities, assessment, answer key, and differentiated content.
Data Exchange	n8n item / JSON-style fields	Carry prepared values and generated outputs between nodes.
Output Layer	Google Docs	Create the combined editable lesson-plan document.
Notification	Gmail	Request missing information and support delivery of the document link.
Version Control	GitHub / project repository	Store workflow export, prompts, diagrams, documentation, and revisions.

8.2 Component Diagram

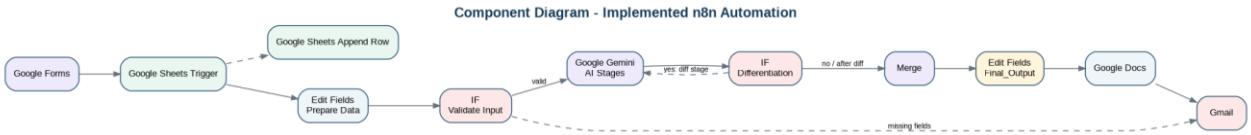


Figure 14. Component Diagram - implemented n8n nodes and connected Google services.

8.3 Deployment Diagram

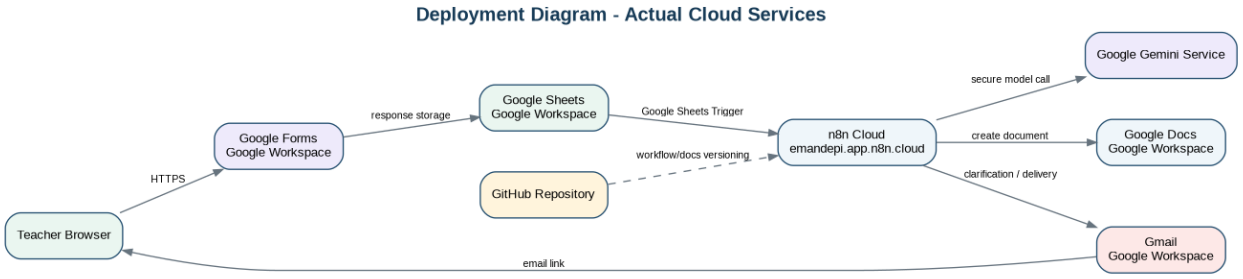


Figure 15. Deployment Diagram - n8n Cloud, Google Workspace services, Google Gemini, and repository.

8.4 Integration Security

- Store Google Workspace and Google Gemini credentials in n8n credentials or secure workspace configuration.
- Do not place API keys, OAuth tokens, or credentials in GitHub, screenshots, documentation, or exported examples.
- Send only the lesson-level information required by the relevant Google Gemini generation stage.
- Validate required input before AI calls and keep clarification/error routes separate from successful output delivery.
- Use the minimum required Google Sheets, Google Docs, and Gmail permissions for the connected project accounts.

9. Additional Deliverables

9.1 API Documentation

The implemented system uses n8n connectors for Google Sheets, Google Gemini, Google Docs, and Gmail. The connector-level design below reflects the supplied workflow documentation. The exact Google Gemini model variant is not specified in the provided implementation materials.

Connector / Integration	Implementation Design
Google Sheets Trigger	Reads a newly stored Google Forms response and starts the n8n workflow with the submitted lesson fields.
Google Sheets Append Row	Stores/records the teacher input data in Google Sheets for logging/record purposes.
Google Gemini Chat Model	Receives stage-specific prompts plus the prepared lesson context and returns generated text for outline, activities, assessment, answer key, and differentiation.
Google Docs	Creates the lesson-plan document and inserts Final_Content into the document body, returning a document reference/link.
Gmail	Sends a clarification request for invalid/missing input and supports sending the generated Google Docs link to the teacher.
Validation / Data Contract	Edit Fields normalizes values; IF nodes validate inputs and differentiation; downstream steps expect non-empty generated output before merge/document creation.

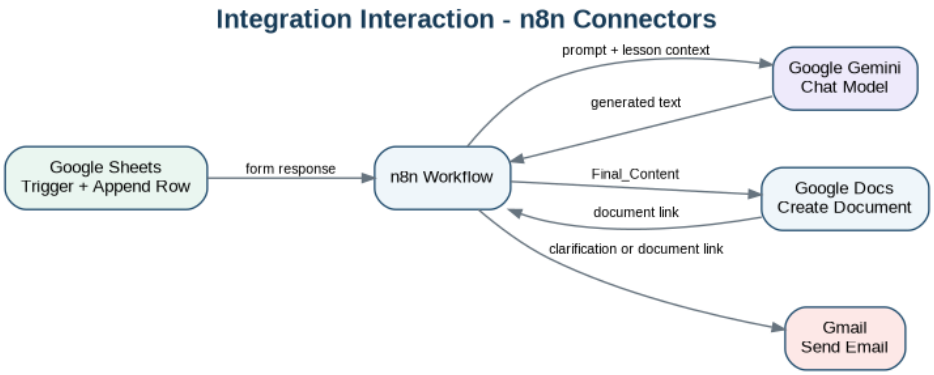


Figure 16. n8n Connector Interaction - Google Sheets, Google Gemini, Google Docs, and Gmail.

9.2 Testing & Validation

Testing should verify the implemented path from Google Forms and Google Sheets through n8n, Google Gemini, Google Docs, and Gmail. The project documentation defines the following success criteria: a new form response starts the workflow, required inputs reach the AI steps, expected lesson sections are combined into Final_Content, a Google Docs document is created and populated, and teacher delivery is complete after the Gmail node is configured and tested.

Test Level	Coverage	Pass Evidence
Node / Unit Testing	Edit Fields preparation; IF validation; differentiation IF; each Google Gemini stage; Merge; Final_Content formatting; Google Docs; Gmail.	Representative input produces the expected node result and required fields.
Integration Testing	Google Forms -> Google Sheets -> n8n -> Gemini -> Merge/Edit Fields -> Google Docs -> Gmail.	Data moves correctly through the complete workflow and the Google Docs document is populated.
User Acceptance Testing	Teacher reviews the Google Docs lesson plan for clarity, relevance, completeness, editability, and differentiation.	Teacher can open, review, and edit the generated document with acceptable corrections.
Failure / Recovery Testing	Missing required fields; invalid differentiation value; AI/service failure; Docs/Gmail connector failure.	Workflow stops or routes safely and provides a clear clarification/error status instead of a misleading final output.
Regression Testing	Re-run previously successful form submissions after prompt or workflow changes.	Existing core behavior remains functional after updates.

9.3 Deployment Strategy

Step	Deployment Activity
1	Connect the Google Form to the Google Sheets response spreadsheet used by the project.
2	Configure the n8n Cloud workflow in the team workspace and connect the Google Sheets Trigger to the response sheet.
3	Configure Google Gemini credentials/model nodes and the stage-specific prompts.
4	Configure Google Docs and Gmail credentials with the minimum required permissions.
5	Run validation, differentiation, end-to-end document-creation, and clarification/delivery test cases.
6	Activate/freeze the validated workflow version for the final demo and preserve the workflow export and documentation in the project repository.



Figure 17. Deployment and change pipeline for the implemented cloud workflow.

10. Design Summary

The final system design now reflects the implemented Automated Teacher Workflow. Google Forms captures teacher input, Google Sheets stores the response and triggers n8n Cloud, Google Gemini generates the lesson sections, IF nodes manage validation and differentiation, Merge/Edit Fields create Final_Content, Google Docs produces the editable lesson plan, and Gmail supports clarification and document-link delivery. The design artifacts in this document are aligned with these implemented services rather than a separate custom application or relational database.

Design Area	Final Design Decision
Primary Actor	Teacher / teaching professional
Project Name	Automated Teacher Workflow
Input & Trigger	Google Forms -> Google Sheets -> Google Sheets Trigger in n8n
Core Platform	n8n Cloud
AI Generation	Google Gemini Chat Model
Core Outputs	Lesson outline; student activities; assessment; answer key; differentiated material when requested; combined Final_Content
Output & Delivery	Google Docs editable lesson plan; Gmail clarification/delivery support
Persistence / Logging	Google Sheets response/log rows; Google Docs final document; n8n execution data
Human Control	Teacher review and editing required before classroom use
Implementation Assets	Shared Google Drive folder containing the exported n8n workflow JSON and supporting project files. Open Shared Google Drive Folder

5. Implementation
(Source Code & Execution)

Introduction

The Automated Teacher Workflow with n8n is an AI-enabled educational automation project designed to reduce repetitive lesson-preparation tasks. The workflow receives instructional information from a teacher, validates required fields, passes the information through AI-powered processing stages, and organizes the generated materials for teacher review.

The workflow is intended to generate a lesson outline, student activities, assessment questions, an answer key, and differentiated learning materials. It integrates n8n with Google Sheets, Google Gemini, Google Docs, and Gmail.

1. Source Code

1.1 Project Overview

The workflow is represented by an exported n8n JSON file. This export records the configured nodes, their parameters, connections, prompts, and data-processing expressions. It provides a source artifact that can be reviewed, imported, and maintained, subject to the availability of valid credentials and compatible service configurations.

1.2 Main Implementation Components

- Input processing: Receives the teacher's subject, grade level, topic, learning objective, lesson duration, and differentiation preference.
- Input validation: Uses an IF node to check whether required fields contain values before AI content generation.
- AI content generation: Uses Google Gemini nodes to generate the lesson outline, student activities, assessment questions, and answer key.
- Workflow logic: Connects processing nodes and uses conditional branches to control execution.
- Output assembly: Combines generated content into a final content field.
- Document generation: Creates a Google Docs document and inserts the generated lesson materials.
- Email notification: Uses Gmail nodes for missing-information and final-output notifications.
- Data logging: Appends teacher input records to Google Sheets.

1.3 Structured and Maintainable Code

The workflow is organized into separately named nodes, including Edit Fields, AI-Lesson outline, Students Activities, AI Assessment, Assessment Answer key, and Create a document. Each node has a defined responsibility. This modular structure makes the workflow easier to understand, troubleshoot, maintain, and extend.

The exported JSON file serves as the main source-code artifact. The workflow also uses n8n expressions to pass values between nodes and reuse teacher-provided information throughout the automation

1.4 Coding Standards and Naming Conventions

Node names generally describe their functions, and expressions are used to retrieve values from incoming items or earlier node outputs. For example:

```
{{ $json.Subject }}
```

This expression retrieves the Subject value from the current input item. Expressions that reference prior nodes allow later AI stages to use previously generated content.

Recommended improvements include standardizing node names, field names, whitespace, output labels, and error messages. File names and documentation should follow a consistent naming convention.

1.5 Modularity and Reusability

The workflow separates educational content generation into distinct modules:

- Lesson outline generation.
- Student activities generation.
- Assessment question generation.
- Answer-key generation.
- Differentiated material generation.
- Final content assembly.
- Document creation and email notification.

This design allows individual prompts or processing stages to be improved without redesigning the entire workflow. Reusing the teacher's original input also avoids asking the teacher to enter the same information repeatedly.

1.6 Security and Error Handling

The exported workflow includes an IF node that checks required teacher inputs. The missing-information branch sends a clarification email instead of proceeding with incomplete input. Several AI nodes also have retry settings configured.

The following improvements are recommended and should be tested before the workflow is described as production-ready:

- Store API keys, OAuth tokens, and passwords in n8n credentials; do not expose secrets in source files.
- Validate required fields and ensure lesson duration is a valid positive number.
- Add clear error handling for failures involving Google Gemini, Google Sheets, Google Docs, and Gmail.
- Define a fallback path for repeated AI or integration failures.
- Avoid hardcoded recipient email addresses where configurable teacher contact information is available.
- Prevent the workflow from sending an incomplete or misleading final document.
- Review the differentiation condition to ensure that it tests the intended preference rather than comparing a field with itself.

1.7 Deliverables

The expected source-code deliverables are:

- The exported n8n workflow JSON file.
- The workflow diagram.
- Documentation describing the main nodes and their connections.
- The AI prompts used for lesson generation and assessment.
- Configuration instructions for the required integrations.
- Documentation of validation, error handling, and fallback behavior.

> The workflow export should be reviewed and tested before it is treated as the final implementation.

2. Version Control & Collaboration

2.1 Version Control Repository

GitHub is recommended as the central repository for the project source code and documentation. A shared repository can hold the n8n workflow export, AI prompts, technical documentation, testing evidence, and implementation updates. The supplied files do not establish that a GitHub repository has already been created, so the repository URL should be added only after creation.

Proposed Repository Structure

```
automated-teacher-workflow/
├── README.md
├── workflow/
│   └── automated_teacher_workflow.json
├── docs/
│   ├── workflow_diagram.png
│   ├── implementation.md
│   └── configuration.md
├── prompts/
│   ├── lesson_outline.txt
│   ├── student_activities.txt
│   ├── assessment.txt
│   ├── answer_key.txt
│   └── differentiated_material.txt
├── tests/
│   └── test_cases.md
└── .gitignore
```

This structure separates workflow source code from documentation, prompts, and testing records, making the project easier to review and maintain.

2.3 Branching Strategy

A feature-branch workflow is recommended so that team members can work on separate components without directly changing the stable version.

- **main:** The reviewed and approved project version.
- **develop:** An integration branch for changes before they are promoted to main.
- **feature/workflow-updates:** Changes to n8n nodes, connections, and workflow logic.
- **feature/ai-prompts:** Improvements to AI prompts and generated educational content.
- **feature/documentation:** README files, configuration instructions, and technical documentation.
- **fix/validation-errors:** Corrections to validation and conditional logic.

Team members should commit changes to the appropriate branch and submit a pull request before merging significant changes into the stable branch.

2.4 Commit History and Documentation

Commit messages should explain changes clearly and help the team identify when specific modifications were introduced. Each commit should focus on a specific change, and important workflow updates should be accompanied by relevant documentation.

```
feat: add teacher input validation

feat: generate lesson outline using Gemini

feat: generate student activities and assessments
feat: create final lesson document in Google Docs
fix: correct differentiation condition

fix: improve missing-input notification
docs: add workflow setup instructions

test: document validation and failure scenarios
```

Pull requests should describe the purpose of the change, affected workflow nodes, tests performed, and known limitations.

2.5 Continuous Integration and Quality Checks

Automated checks may be introduced where suitable. Recommended checks include:

- Validate that the exported workflow JSON is syntactically valid.
- Review changes to workflow nodes and connections.
- Verify that required documentation and project files are included.
- Test valid inputs, missing fields, and failure scenarios.
- Import updated workflow versions into a test environment before using them in the final environment.

Because the workflow depends on external services, successful execution also requires valid credentials and appropriate access. CI/CD pipelines are optional improvements and should only be reported as implemented if they have actually been configured and tested.

2.6 Repository Security and Backup

The repository should not contain API keys, OAuth tokens, passwords, or other sensitive credentials. The .gitignore file should exclude local secrets, temporary files, and unnecessary artifacts. Exported workflows should be reviewed for sensitive configuration details before sharing.

A reviewed workflow export should be retained at important milestones so that the team can restore a previous configuration if an update causes problems.

2.7 Deliverables

- A GitHub repository containing project source files.
- An organized repository folder structure.
- A documented branching strategy.
- A meaningful commit history.
- Pull requests or review records for significant changes.
- A README with installation and execution instructions.
- A documented final version of the n8n workflow.

These practices support collaboration, traceability, maintainability, and safer project updates.

3. Deployment & Execution- README File

README File

The Readme File Provides Project Overview and The Instructions Required to Install, configure, execute, and deploy the Automated Lesson Plan Generation System.

It is intended for developers, project evaluators, administrators, and teachers who need to understand how the system operates.

3.1 Installation Steps

Install the Required Software

There are Two installation methods, first Cloud version of n8n and the second id self Hosted like Docker Environment.

We will use n8n cloud version as the primary workflow automation platform, and create an account on, then build the workflow or import it into n8n.

3.2 System Requirements

3.2.1 Hardware:

CPU	Minimum 2 Cores (4 Cores recommended
RAM	Minimum 2-4 GB RAM (8 GB+ recommended).
Storage	5 GB storage Free.
Internet	Stable connection, High speed connection

3.2.2 Software Requirements

- Operating System Windows 10/11, Linux, or mac
- Node.js or Docker as n8n Cloud
- Google Account (Valid Gmail Account)
- Google Forms
- Google Sheets
- Google Gemini API
- Google Docs
- Web Browser

3.2.3 Software Dependencies

The system depends on:

n8n cloud account	Responsible for connecting all system components and executing the automation workflow.
Google Gemini API	As AI Engine Responsible for generating lesson outlines, student activities, assessments, answer keys, and differentiated educational content.
Google Workspace APIs	Google Forms (Input Interface) Google Sheets (Logging Database) Google Docs (Output Document Builder) Gmail (Notification & Delivery System)
Web Browser	Updated Google Chrome, Firefox or Edge

3.3 Configuration Instructions

Before executing the workflow, all external services must be configured correctly.

1. Create a Google Forms as it acts as the main input interface for teachers.
2. Create Google Sheets to Store teacher's submissions and trigger the n8n workflow with newly added rows and start the lesson generation process.
3. Google Gemini API Configuration
    - Step 1: Obtain a Gemini API Key
    - Step 2: Configure Gemini in n8n

3.4 Execution Guide

- **Node 1: Google Form (Teacher Imputes) :** The Teacher submits a Google Form containing key metadata: Subject, Grade Level, Topic, Learning Objective, Lesson Duration, and Differentiation Preference.
- **Node 2: Google Sheets Trigger:** The form submission triggers the n8n automation engine to check the connected spreadsheet for new rows add, when a new teacher request is detected, the data is passed to the next node.
- **Node 3: Data Preparation.** Edit and recreates and renames form fields for internal processing
- **Node 4: Validation Input (IF):** The system checks and ensure that all required information is available before generating the lesson plan like (The teacher's email, address is provided, the subject is specified, the grade level is selected, the lesson topic is provided, Learning objectives are available, Lesson duration is a positive number.)
    - IF Execution Conditions is Valid The workflow continues to the AI generation.
    - IF Execution Conditions are invalid the workflow moves to the clarification email stage by Triggering a Gmail notification to the teacher requesting clarification for missing fields.
- **Node 5: AI Lesson Outline:** Generate the lesson structure lesson outline.
- **Node 6: AI Student Activities:** Generate interactive educational activities aligned with the learning objectives
- **Node 7: AI Assessment:** Automatically generate A structured evaluation materials (MCQs, short answer, application questions) to evaluate student understanding aligned with the learning objectives.
- **Node 8: AI Answer Key:** Generate an accurate answer key and explanations corresponding to the assessment questions.
- **Node 9: Differentiation (IF):** Determine whether the teacher requested differentiated learning activities.
    - IF True: The workflow activates the differentiated content.
    - IF False: The workflow skips the differentiation.
- **Node 10: AI Differentiated Version:** Generate adapted activities to support students with different learning needs.
- **Node 11 – Merge (Combine Outputs):** Combine the generated components into One structured lesson plan object containing all generated sections like (Lesson Outline, Student Activities, Assessment Questions, Answer Key)
- **Node 12: Edit Fields:** The system Dynamically Prepare and organizes the lesson content into a consistent format follow a standardized structure.
- **Node 13: Google Docs Creation:** The workflow Dynamically sends the formatted content to Google Docs; the system creates a new document containing the generated the completed lesson plan.
- **Node 14: Google Sheets (Append Row):** The workflow mapping the teacher's input fields information in a logging spreadsheet, so every submission is recorded even when validation fails.
- **Node 15: Gmail Request Clarification:** The input validation stage identifies incomplete or invalid information; the system here notifies the teachers when required information is missing.
- **Node 16: Gmail Sent to Teacher:** After the Google Docs document has been created, the workflow Sends a Gmail email to the teacher containing the completed final lesson plan and direct link to the Google Doc generated successfully

Execution Guide (Running Locally)

1. Start n8n Cloud account.
2. Open the Workflow Editor
3. Import the project workflow JSON.
4. Open the imported workflow
5. Verify that all nodes are imported successfully and every node shows connected credentials (no red warnings).
6. Configure Google credentials.
7. Configure the Gemini API credentials.
8. Verify the Google Sheets connection.
9. Test the workflow by Click Execute Workflow (manual test) or toggle Active to run automatically.
10. Submit a test request using Google Forms.
11. Verify successful document creation and email delivery.

3.5 API Documentation

The system uses external APIs rather than requiring a custom backend API.

The main APIs are:

API	Purpose	Authentication
Google Sheets API	Retrieve teacher requests and maintain submission records.	OAuth 2.0
Gemini API	Generate lesson outlines, classroom activities, assessment questions, answer keys, and differentiated activities.	API Key
Google Docs API	Create and edit documents	OAuth 2.0

Executable Files & Deployment Link

Executable Files

- This system is web-based automation solution built using n8n workflow and integrated cloud services.
- The workflow is named Automated Teacher Workflow. Json file.
- RAR Folder Contain all documents and executable files

Deployment Links

1. Teacher Input Form
   - https://docs.google.com/forms/d/e/1FAIpQLSese_DKFHP6xJmFg9HCyhLqn9P6r9b_mopQ4nipkbrWmgCVCw/viewform
2. n8n Platform
   - https://emandepi.app.n8n.cloud/workflow/8w9bjCQRTa98ljvH
3. Generated Lesson Plan
   - https://docs.google.com/spreadsheets/d/12NtXNeaw6yFkQJYLQdJzBFHiLA_YR-VLuoeddvgvMRo/edit?resourcekey=&gid=840199554#gid=840199554

6. Testing & Quality Assurance

1. Overview

The existing workflow was reviewed and manually tested node by node. Testing covered data mapping, conditional routing, AI-generated lesson content, document creation and email delivery. A test-data end-to-end execution was also performed.

The main functional issue was that the original Google Docs route displayed Markdown syntax as plain text. A new path was tested using Markdown-to-HTML conversion, a JavaScript multipart request body, and a Google Drive HTTP upload. The resulting document displayed formatted content. Gmail was updated to use the new document link.

The Append Row node was also tested separately using a QA sheet; the tester confirmed that a row was added. It was disabled in the client handoff template to avoid unnecessary writes or duplicated records.

2. Test scope and results

The following statuses reflect observed manual testing, the tester's confirmations, and configuration review. "PASS by tester" indicates user-confirmed behavior rather than an independently captured output. Some AI content-quality issues and live-deployment checks remain open.

Test case / Component	Status	Observed result / Note
TC-1 Google Sheets Trigger (Fetch Test Event)	PASS	34 sample records; live new-row activation not proven.
TC-2 Edit Fields	PASS	Six required fields mapped for one input item.
TC-3 If - valid route	PASS	Complete input routed to true branch.
TC-4 AI-Lesson outline	PASS	45-minute lesson outline generated; scope concerns noted.
TC-5 Students Activities	PASS	Five activities generated; some advanced/out of scope.
TC-6 Merge1	PASS	Activities and outline retained in distinct fields.
TC-7 AI Assessment + retests	OPEN CONTENT ISSUE	Answer key removed; multiple-choice quality was variable across runs.
TC-8 Merge2	PASS	Assessment, activities, lesson outline preserved.
TC-9 Assessment Answer key	PASS	Numbered answers generated; upstream item ambiguities persist.
TC-10 Edit Fields2	PASS	Answer key mapped to separate answer_key field.
TC-11 Merge3	PASS	Four content fields retained in merged item.
TC-12 AI Differentiated material	PASS	Advanced material generated; some incorrect debugging instructions.
TC-13 Edit Fields1	PASS	Five sections assembled in Final_content.
TC-14 Create a document (legacy route)	PASS	Test Google Doc created; route later disconnected.
TC-15 Update a document (legacy route)	PASS	Update accepted; document body verified separately.
TC-16 Gmail Send (legacy route)	PASS	SENT message ID returned.
TC-17 Email link access (legacy route)	PASS by tester	Email received and link opened for test account.
TC-18 If - missing Topic	PASS	Blank Topic routed to false branch.
TC-19 Markdown to HTML	PASS	HTML field generated; five sections preserved (shared JSON).
TC-20 Full main-path integration	PASS	User screenshot showed green nodes through Gmail.

Test case / Component	Status	Observed result / Note
TC-21 Code - HTML multipart prep	PASS	uploadBody and contentType returned (shared JSON).
TC022 HTTP Request - HTML conversion	PASS	Google Docs file id, MIME type and URL returned (shared JSON).
TC023 Formatted Google Doc	PASS	Rendered headings, bold, lists; latest link reconfirmed.
TC024 Gmail link preview	PASS	Expression resolved latest HTTP Request webViewLink.
TC-25 Formatted-route wiring	PASS	HTTP Request connected to Gmail; legacy path detached.
TC-26 New Gmail route	PASS	Tester confirmed new path operates; no dedicated JSON screenshot.
TC-27 Missing-information Gmail	PASS	SENT label and message ID returned on invalid-input path.
TC-28 Append row (separate test sheet)	PASS	Mapped response added; tester confirmed row in test sheet.

3. Issues and remaining limitations

D-01 | Append-row duplication risk

A side-effecting append stage previously targeted the form data pipeline. It was safely tested on a separate QA sheet. Client template keeps the node disabled to avoid unintended writes.

D-02 | MCQ ambiguities / Python correctness

AI Assessment retests removed the duplicated answer key, but examples included valid alternatives in more than one option and uppercase AND. These errors are generated-content quality issues and remain a regression risk.

D-03 | Differentiation beyond lesson scope

Advanced output sometimes introduced loops, functions, exceptions and other techniques inconsistent with a 45-minute Python basics objective. Educational quality requires human review.

D-04 | Google Docs raw Markdown

Legacy Google Docs Update inserted Markdown markers (### and **). Corrected flow converts Markdown to HTML then uses Drive multipart upload/conversion. Visual formatting was checked.

D-05 | Fixed test resources and recipient

Source exported with tester Gmail address, source sheet, pinned row, test folder and credential references. Client template removes references and requires client-specific configuration.

D-06 | Live trigger and recipient access

Manual Fetch Test Event and pinned-input end-to-end tests do not prove unattended new-row triggering. Document link opened for the tester; access for other recipients still depends on Drive sharing.

4. Client setup required

The delivered importable JSON is a template with client-specific references removed. It was not re-executed after the placeholder values were inserted. Before activating the imported workflow, complete these steps:

1. Connect client-owned credentials. Google Sheets Trigger, Sheets, Google Drive, Gmail, and Gemini/AI model.

2. Select spreadsheet and response tab. Replace placeholder spreadsheet ID and tab ID in trigger; preserve column names/spaces.

3. Set output folder. Replace REPLACE_WITH_DESTINATION_FOLDER_ID in Code in JavaScript.

4. Set email recipients. Change CONFIGURE_RECIPIENT@example.com on both Gmail nodes; consider adding a teacher-email field.

5. Keep append node disabled. Separate append test passed, but appending is unnecessary for the main flow.

6. Verify document sharing. Check access as the actual recipient, not only as the connected Google account.

7. Run genuine end-to-end validation. Submit a new Form response with the trigger active and check document + email.

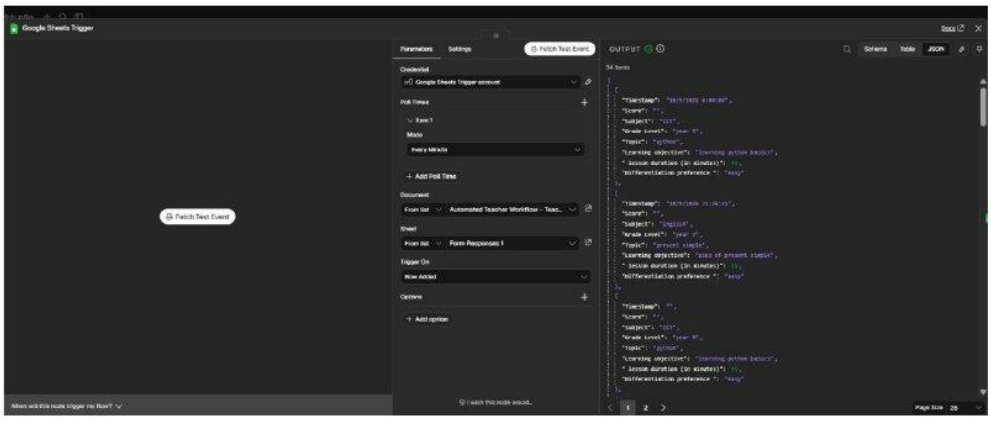
8. Inspect educational material. Review questions, answer key, and differentiated content for accuracy and age appropriateness.

Additional checks: keep the workflow inactive until the real trigger, Google Drive sharing permissions, valid inputs, missing-field handling and educational correctness have been confirmed in the client environment.

5. Screenshot evidence

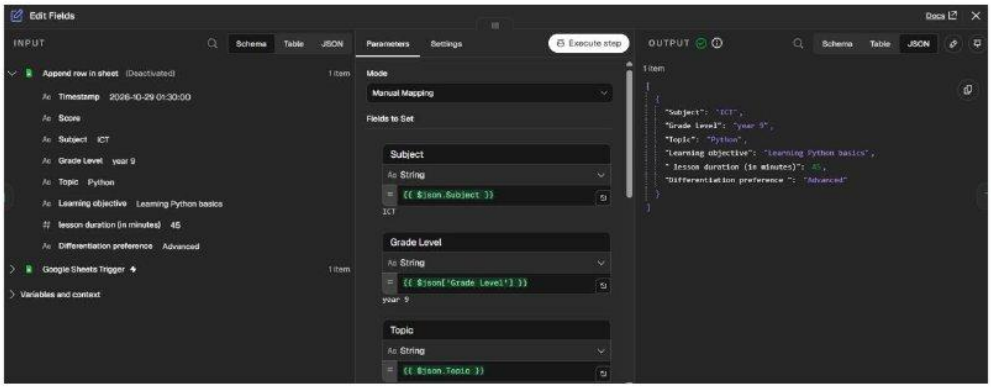
26 screenshots from the testing session are included below. The image files are also provided separately in the Evidence folder. Some email identifiers or document links have been redacted.

TC-1 - Google Sheets Trigger Fetch



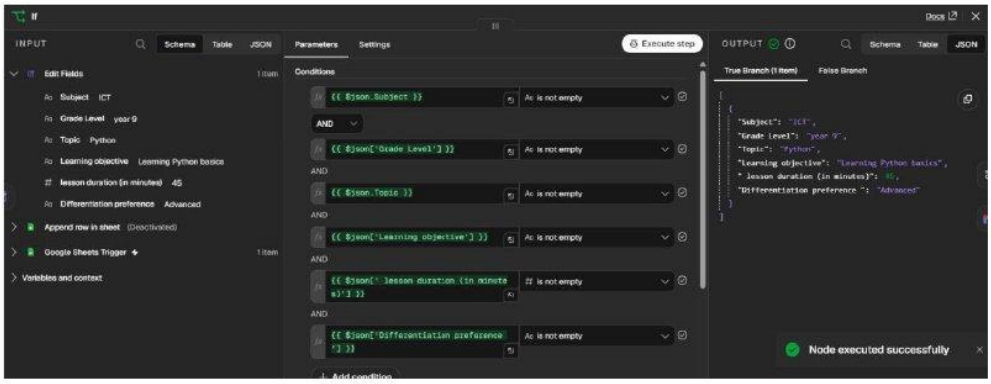
Source: screenshot from manual workflow testing. Redactions applied where needed.

TC-2 - Edit Fields Mapping



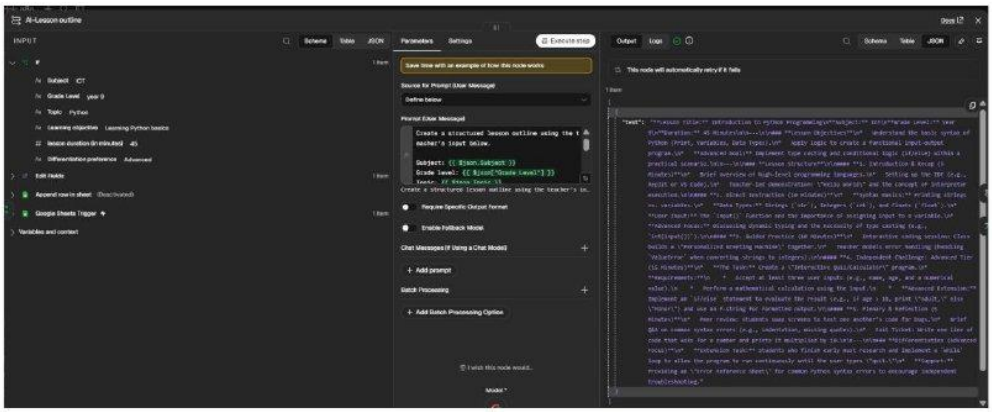
Source: screenshot from manual workflow testing. Redactions applied where needed.

TC-3 - If Valid True Branch



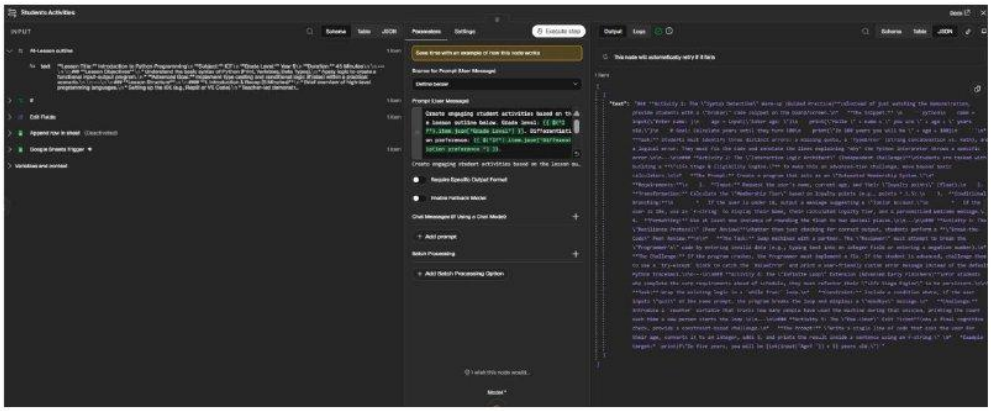
Source: screenshot from manual workflow testing. Redactions applied where needed.

TC-4 - AI Lesson Outline



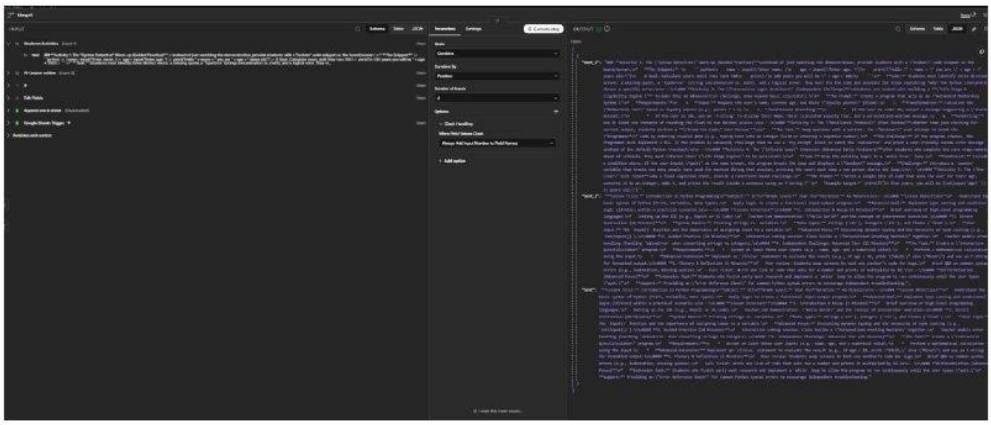
Source: screenshot from manual workflow testing. Redactions applied where needed.

TC-5 - Student Activities



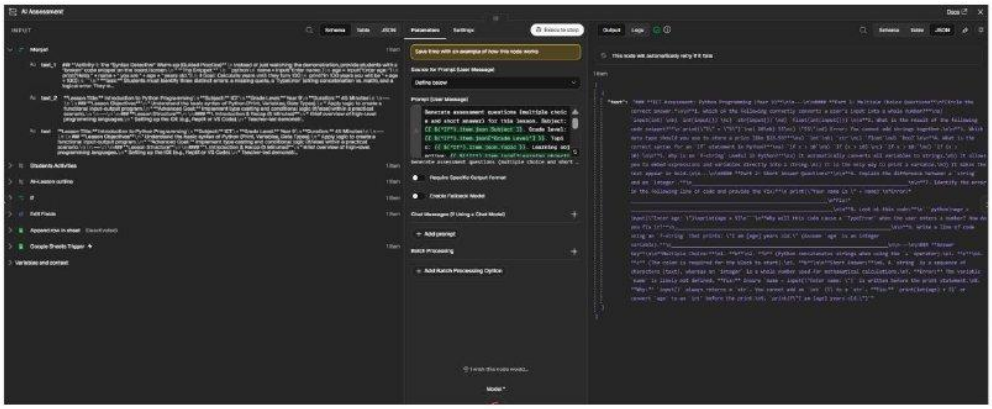
Source: screenshot from manual workflow testing. Redactions applied where needed.

TC-6 - Merge1



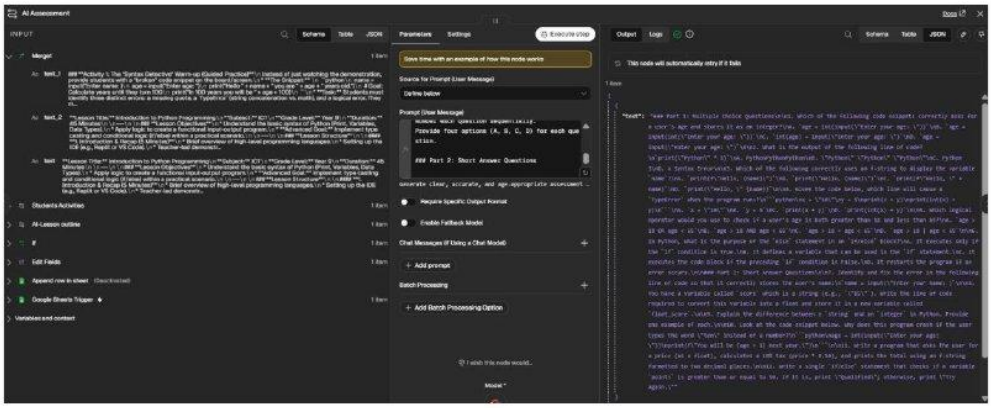
Source: screenshot from manual workflow testing. Redactions applied where needed.

TC-7 - AI Assessment Initial Problem



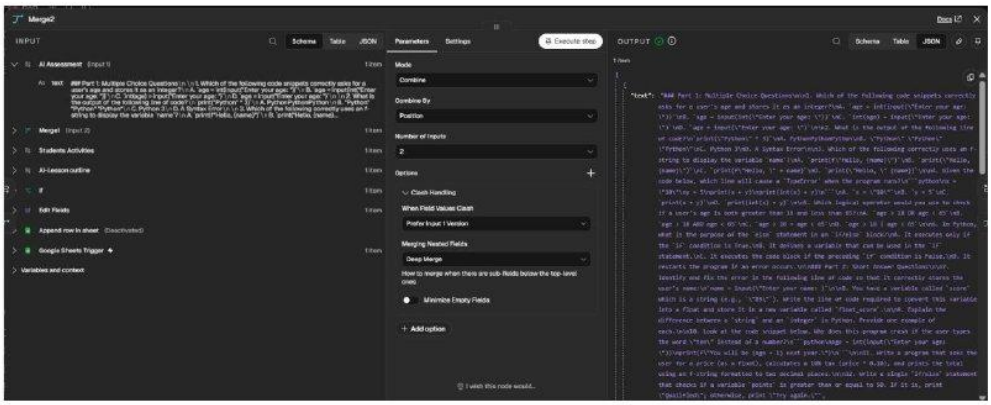
Source: screenshot from manual workflow testing. Redactions applied where needed.

TC-7 - AI Assessment Retest 3



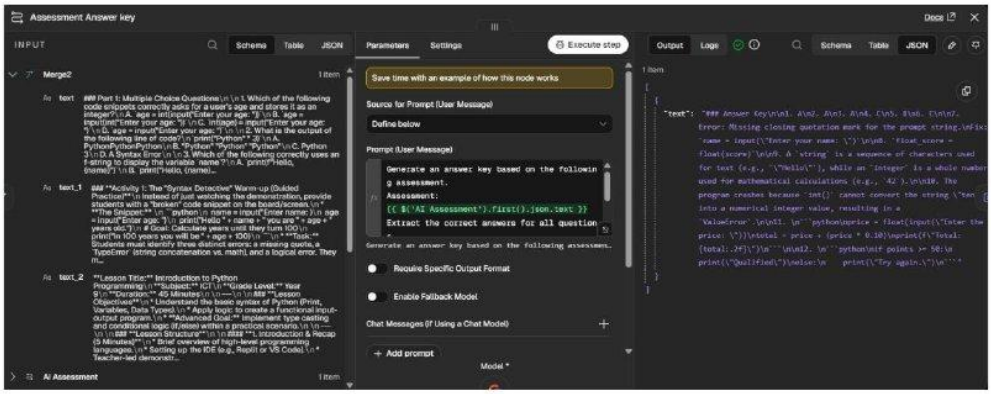
Source: screenshot from manual workflow testing. Redactions applied where needed.

TC-8 - Merge2



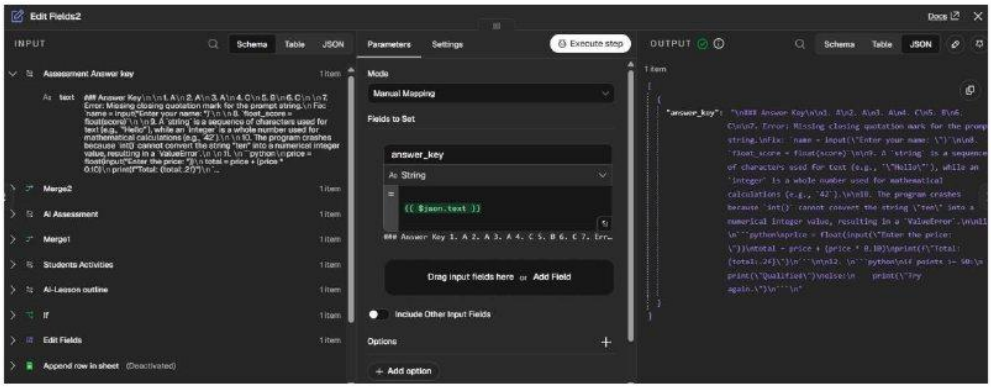
Source: screenshot from manual workflow testing. Redactions applied where needed.

TC-9 - Assessment Answer Key



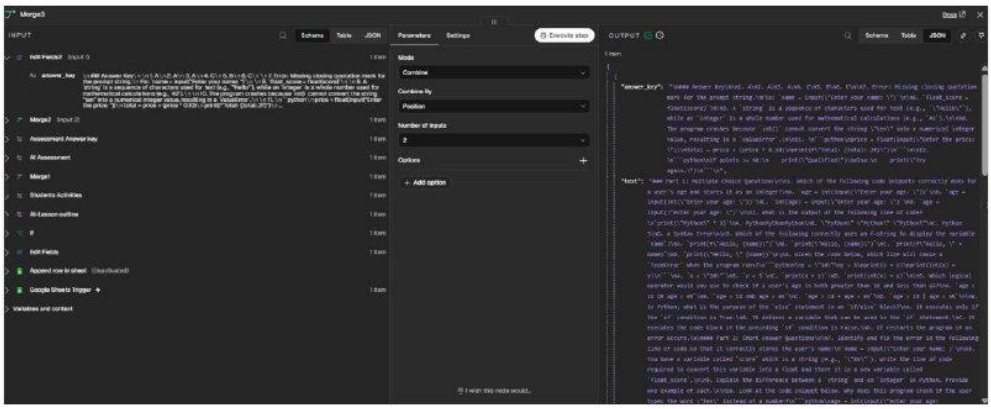
Source: screenshot from manual workflow testing. Redactions applied where needed.

TC-10 - Edit Fields2



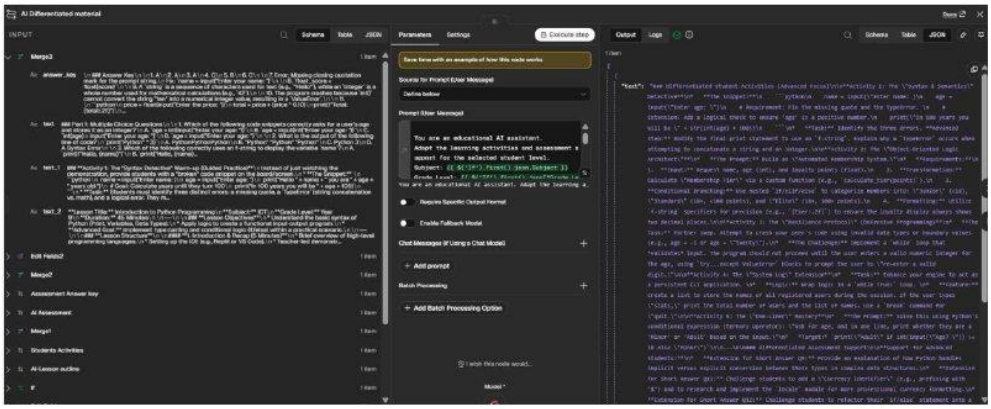
Source: screenshot from manual workflow testing. Redactions applied where needed.

TC-11 - Merge3



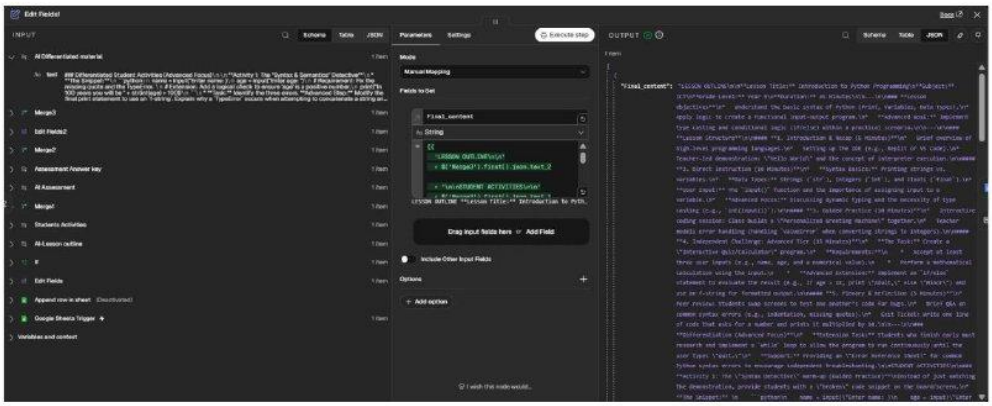
Source: screenshot from manual workflow testing. Redactions applied where needed.

TC-12 - Differentiated Material



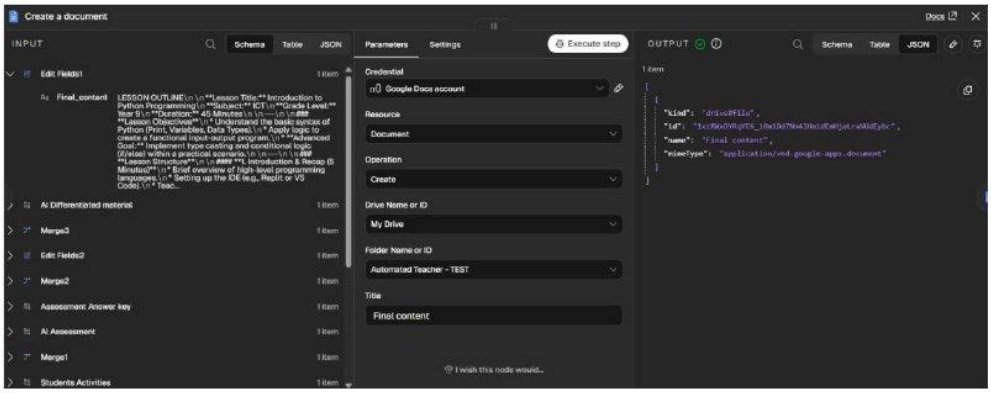
Source: screenshot from manual workflow testing. Redactions applied where needed.

TC-13 - Final Content Assembly



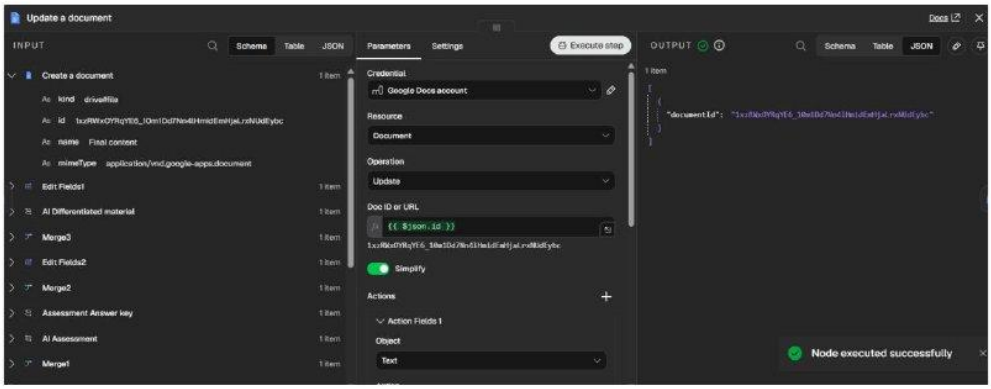
Source: screenshot from manual workflow testing. Redactions applied where needed.

TC-14 - Create Document



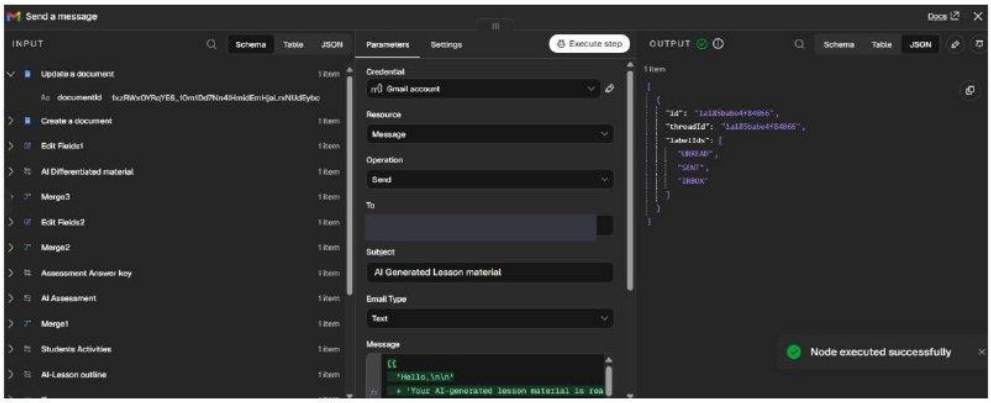
Source: screenshot from manual workflow testing. Redactions applied where needed.

TC-15 - Update Document



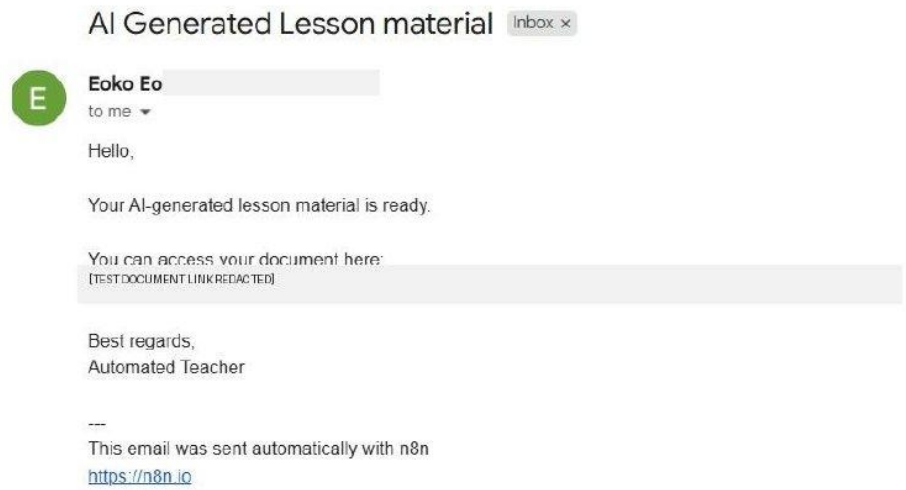
Source: screenshot from manual workflow testing. Redactions applied where needed.

TC16 - Gmail Operation



Source: screenshot from manual workflow testing. Redactions applied where needed.

TC-17 - Email Received



AI Generated Lesson material Inbox ×

E

Eoko Eo

to me

Hello,

Your AI-generated lesson material is ready.

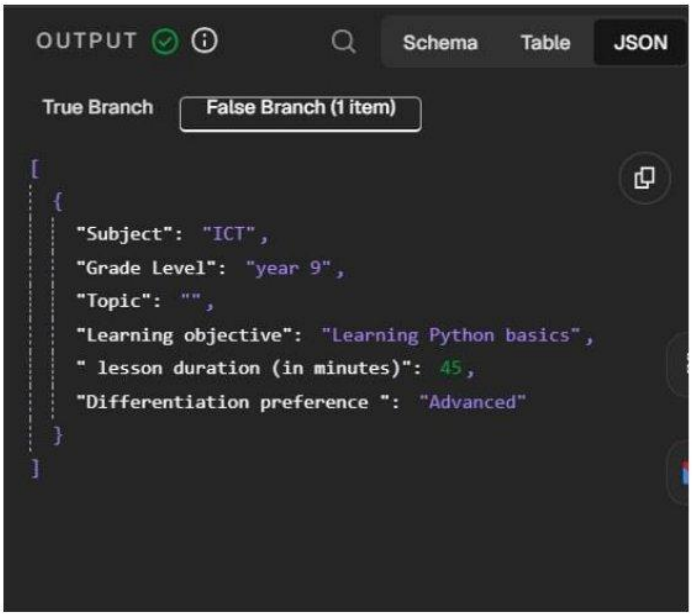
You can access your document here:
[TEST DOCUMENT LINK REDACTED]

Best regards,
Automated Teacher

This email was sent automatically with n8n
https://n8n.io

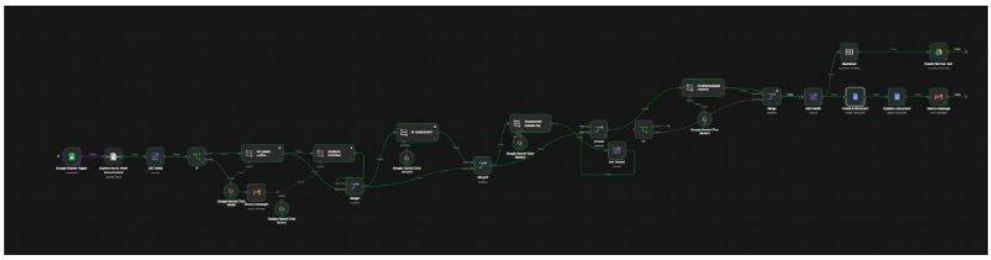
Source: screenshot from manual workflow testing. Redactions applied where needed.

TC-18 - If Missing Topic



Source: screenshot from manual workflow testing. Redactions applied where needed.

TC-19 - Integration Execution



Source: screenshot from manual workflow testing. Redactions applied where needed.

TC-20 - Formatted Document

LESSON OUTLINE

Lesson Plan: Introduction to **Python Programming**

Subject: ICT
Grade Level: Year 9
Topic: Python Basics (Variables, Data Types, and Logic)
Duration: 45 Minutes
Differentiation: Advanced (Focus on procedural abstraction and algorithmic problem solving)

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1. Starter: Logical Thinking Challenge (5 Minutes)

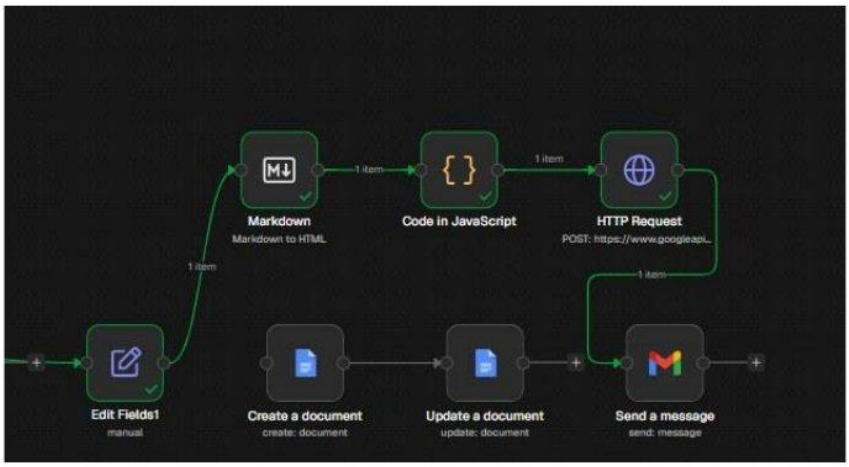
- **Activity:** Present a "pseudocode" snippet on the board representing a common algorithm (e.g., swapping two variables or a simple loop).
- **Task:** Students must predict the output without executing the code.
- **Goal:** Activate prior knowledge of computational logic.

2. Direct Instruction: Core Concepts (10 Minutes)

- **Topics:**
  - Dynamic vs. Static typing in Python.
  - Input/Output handling (input(), print()).
  - Type casting (int(), str(), float()).
  - Introduction to Boolean logic (if, elif, else).

Source: screenshot from manual workflow testing. Redactions applied where needed.

TC-21 - New Final Connections



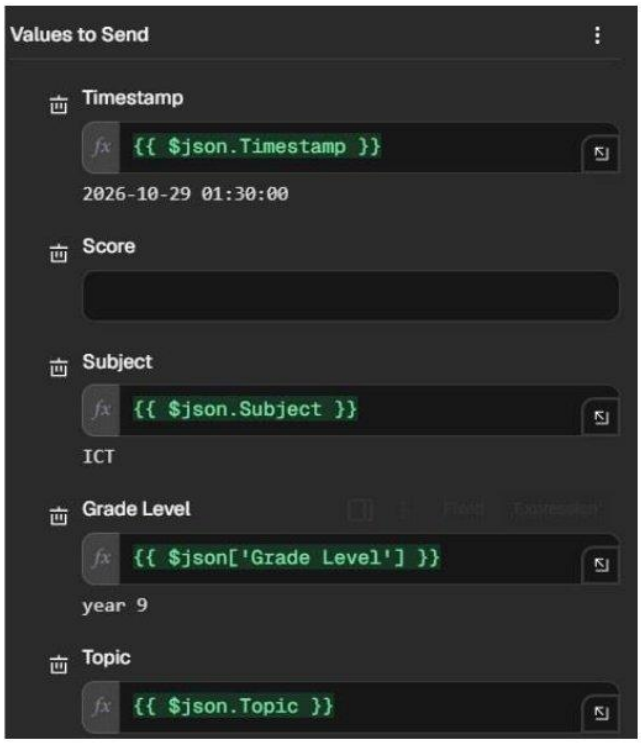
Source: screenshot from manual workflow testing. Redactions applied where needed.

TC-22 - Missing Information Email Output



Source: screenshot from manual workflow testing. Redactions applied where needed.

TC-23 - Append Mapping To Test



Source: screenshot from manual workflow testing. Redactions applied where needed.